

ITCS213 Computer Organization and Architecture

Class: 1-2025

Sec 2 Wed 9:00-12:00 IT312 (Aj. Joke)
Sec 3 Thu 9:00-12:00 IT312 (Aj. Joke)
Sec 1 Thu13:00-16:00 IT311 (Aj. Peter)

Unit 1: Introduction to **Computer Architecture**

Lecturer: Dr. Peter M.D. Scully
Faculty of ICT, Mahidol University
peter.scu@mahidol.edu

Slides adapted from Subject Coordinator:
Dr. Kanoksak Wattanachote (Ph.D) (金功勳 博士)
kanoksak.wat@mahidol.ac.th

Part

Course Logistics

The course outline, the assessment, objectives and weekly logistics.

Topics Outline | Quizzes | Lab Exercises | Exams

Course Logistics — ITCS 213 – Computer Organization and Architecture

Number of Credits	3
Lecture-Lab-Self Study	(3-0-6)
Type of Course	Compulsory Course
Program	Bachelor of Science Program in ICT (International Program) (B.Sc.)
Session	1st Semester

Lecture Time	(Room IT311) Fri 13:00 PM -- 16:00 PM

Lecturer(s)	Dr. Peter M. D. Scully (Weeks 1-8) – Computer Organization and Architecture + Assembly (Part 1) (Weeks 9-15) – Processor Design, Memory Management + Assembly (Part 2)
	Email + MS Team: peter.scu@mahidol.ac.th
Subject Coordinator:	Dr. Kanoksak Wattanachote Email + MS Team: kanoksak.wat@mahidol.ac.th

Course Logistics — Lecturer



Dr. Peter M. D Scully, PhD

Instructor of Computer Science
Faculty of ICT, Mahidol University

peter.scu@mahidol.ac.th

My name:
- **First Name:** **Peter**
- **Family Name:** **Scully**

Common Name / Short name :
- **Short name:** **Peter / Pete / Aj. Peter**

Education
- Ph.D. (Computer Science), Aberystwyth University, United Kingdom, 2016
- BSc. (Computer Science), University of Wales, Aberystwyth, United Kingdom, 2011

Teaching of Data Science, IOT and Cybersecurity Areas:
- 2026 - Embedded Systems and Internet of Things, BSc ICT Program, Mahidol University
- 2026 - Data Science for Cybersecurity, MSc CY Program, Mahidol University
- 2026 - Data Science, Introduction and its Applications, BSc ICT Program, Mahidol University
- 2025 - Embedded Systems and Internet of Things, BSc ICT Program, Mahidol University
- 2025 - Data Science, Introduction and its Applications, BSc ICT Program, Mahidol University
- 2024 - Edge Computing and Internet of Things, MSc CS Program, Mahidol University
- 2024 - Data Science Essentials and Applications*, BSc Creative Technology, Mahidol University.
- 2022-23 - Data Science for Health Informatics, MSc Health Sciences Innovation, Mahasarakham University.
- 2015-2018 - Data and Software Engineering - TAP Grad Program, **Morgan Stanley (Fortune 100), India**
- 2015-2016 – Ubiquitous Computing, BSc Aberystwyth University, **United Kingdom (*IOT App for the Zombie Apocalypse*)**
- 2014-2015 - Programming for Scientists (Python) – MSc Interdisciplinary Scientists, Aberystwyth University, **United Kingdom**

Research in Cybersecurity & Health Informatics & Data Acquisition in Research Systems (Signal Data):
- PhD Research in Cybersecurity. Research Assistant (PhD) with **Airbus Innovation Labs (Fortune 500), United Kingdom.** Head of Cyber Security Research Labs and CISO Industry Advisors.
  - Self-healing architectures (AI) for industrial control networks. (Stuxnet, Botnets, and similar).
  - Machine learning, Anomaly detection, recovery solution generation, decentralized network security.
  - More info: petescully.co.uk

Research area processes:
- Intelligent Systems, Edge Computing, Signal Data, Time Series Modeling, Multimodal Models, Visual-Recognition, Policy Analysis.

Recent Research Activities / Achievements (2025-26)
- **Principal Investigator.** Science Research and Innovation Fund (SRIF) : University Fundamental Fund 2026 (750,000 THB) CARBONCOW - Improving the Accuracy and Volume of In vivo Methane Quantification: Pathways towards Carbon Credits via Methane Mitigation in Livestock in Thailand & South-east Asia. Thailand.
- **Program Committee & Reviewer, MIWAI 2025** - 18th Multi-Disciplinary International Conference on Artificial Intelligence on December 3-5, 2025, Vietnam.
- **Reviewe**r for the Journal of Healthcare Informatics Research (I.F. 5.1), International Journal of Geoinformatics (I.F. 2.2)
- **Best Paper Award,** “Intelligent Safety-Redundant IoT-Edge System for Indoor Gas Leakage Detection”, Ubi-Media 2025, The Thirteenth International Conference on Ubi-Media Computing, Published by Springer in Communications in Computer and Information Science (CCIS) series. January 2025.

Course Logistics — Learning Outcomes

1. To **understand** the **basic concepts** of **computer organization and architecture**.
2. To **understand** the **principle components** of computers, and **how each component communicates** among each other both internally and externally.
3. To understand computer systems with **single-core processors** and **multiple-core processors**.
4. To **apply** the **concept and practices** of computer organization and architecture with awareness of **constraints to solve problems** through class exercises, assignments and examinations.

Course Logistics — Course Outline

Weeks 01–08

(Part 1)

CPU Architecture detail and begin Assembly Language.

Weeks 09–15

(Part 2)

More Assembly, more logic and detail on how computer systems work.

NO	TOPIC	LECTURE
	Computer Organization and Architecture + Assembly Language (Part 1)	
1	**Computer Systems, Architecture Fundamentals, and Data Representation**	**3**
2	ALU, Instruction Set Architectures, and Assembly Fundamentals	3
3	Instruction Pipelining and Hazards	3
4	Arithmetic Algorithms, Hazard Resolution, and MIPS Assembly	3
5	Memory Hierarchy, Cache, and MIPS Data Transfer	3
6	Integer Representation, Bitwise Logic, and Shift Operations	3
7	I-Type Instructions and MIPS Immediate Operations	3
8	I-Type Instructions and Datapath, MIPS Multiply/Divide	3
	Midterm Exam (#1-8) 2 hours on FinEx	
	Processor Design, Memory Management + Assembly Language (Part 2)	
9	R-Type vs. I-Type Datapath and Control Flow	3
10	Control Flow, Arrays, and Memory Addressing	3
11	Floating-Point Arithmetic and Processor Design Introduction	3
12	Floating-Point Arithmetic and Datapath Design	3
13	Datapath Components and Clocking	3
14	Memory Management, Performance Analysis, and Superscalar Processors	3
15	*Applied Concept Project Report (Course Capstone)*	3
	Final Examination (#9-14) 2 hours on FinEx	

Course Logistics — Class Format *(Initial Plan // Best Intentions)*

Time (PM)	Activity
13:00 – 13:10	Announcements / Attendance
13:10 – 13:50	**Lecture**
	Short Break
14:10 – 14:50	**Lecture / Activity**
	Short Break
15:10 – 15:50	**Lecture / Activity**
15:50 – 16:00	*Break / Vacate the lab.*

(roughly…)

In Class Activities / Exercises:

- **Bring a laptop to the room**. *(From **Week 3**)*
  - *Work on lab exercises in class and after class.*
  - *Quiz after class (often)*

3 hours is a long time for lectures on Friday PM! 🥱

May try AI gamification…

- *Flashcard Search & Destroy*
- *In-class / Live QA Competitions*
- *Lightning Round Calculations*
- *Short Explainer Podcasts*
- *…* 😊

Course Logistics — Course Outline Exercises - Part 1

#	Topics	Lab Exercises (Bring laptop from #3)	Quizzes
Computer Organization and Architecture + Assembly Language (Part 1)			
1	Computer Systems, Architecture Fundamentals, and Data		Quiz 1
2	ALU, Instruction Set Architectures, and Assembly Fundamentals		Quiz 2
3	Instruction Pipelining and Hazards	Lab: Assembly Coding	Quiz 3
4	Arithmetic Algorithms, Hazard Resolution, and MIPS Assembly	Lab: Introduction MIPS Assembly Language	Release Unit 4 Assignment
5	Memory Hierarchy, Cache, and MIPS Data Transfer	In-class Assembly Practice	Unit 4 Assignment Due Release Unit 5 Assignment Quiz 4-5 (Parts 1-2)
6	Integer Representation, Bitwise Logic, and Shift Operations	Lab: Integer Arithmetic	Quiz 6
7	I-Type Instructions and MIPS Immediate Operations	In-class Assembly Practice	Quiz 7
8	I-Type Instructions and Datapath, MIPS Multiply/Divide	In-class Assembly Practice	
Midterm Examination (#1-8) 2 hours on FinEx			

2025 Activity Schedule (Subject to change)

Course Logistics — Assessments

Evaluation Methods:

NO	EVALUATION METHOD	FULL SCORE
1	**Assignments** / Quizzes	**20 %**
2	**Midterm Exam**	**40 %**
3	**Final Exam**	**40 %**
Total		100 %

Grade Score Mapping Criteria

(Subject to change)

Grade		Score
A	>=	80
B+	>=	75
B	>=	70
C+	>=	65
C	>=	60
D+	>=	55
D	>=	50
F	<	50

Course Logistics —Grading Rubric

Rubric Name : Generic In-Class Activities and Assignments							
Evaluation Method : Assignment							
CRITERIA NAME	**MAX LEVEL**	**3 SOPHISTICATED**	**2 COMPETENT**	**1 TRANSITIONAL**	**0 LACKING**	**WEIGHT**	**SCORE**
Technical Accuracy	3	Work is correct, complete, and demonstrates mastery of the concepts.	Work is mostly correct with minor errors.	Work has significant errors but shows some understanding.	Work is incorrect, incomplete, or not submitted.	1	/ 3
Problem Solving	3	Applies logical methods effectively; solves problems clearly and efficiently.	Good approach with minor gaps in logic.	Basic approach but struggles with key steps.	No clear approach or unable to solve the problem.	1	/ 3
Use of Tools	3	Proficient and confident in using required software/tools.	Good use of tools with minimal assistance.	Basic use of tools; requires frequent guidance.	Unable to use tools or complete the task.	1	/ 3
Clarity & Organization	3	Work is well-organized, easy to follow, and professionally presented.	Work is organized with minor clarity issues.	Work is somewhat disorganized or unclear.	Work is messy, confusing, or hard to understand.	1	/ 3
Participation **Marker's Discretion:** (Overall Polish, Creativity, Professional Conduct)	3	Actively engaged, contributes ideas, and collaborates effectively.	Participates but with occasional disengagement.	Minimal participation or relies on others.	Does not participate or disrupts the activity.	1	/ 3
					Total		**/ out of 15**

Unit 1: Introduction to Computer Architecture

Announcement: ITCS213

Unit 1: Introduction to Computer Architecture

Announcements

- No announcements at this time.

Agenda

Part A: Computer Architectures (Introduction)

Part B: Old Computer

Part C: Secondary Storage Architecture

Part D: Primary Storage Architecture

Part E: Processing Unit

Part F: Computer Architectures (Key Diagrams)

Computer Organization Study Process:

How we build understanding — from hardware components to architecture diagrams

Step 1: Computer Hardware

Storage devices, RAM, CPU and multicore processors

Step 2: Organization vs. Architecture

Designer vs. architect perspectives, structure and function

Step 3: Essential Diagrams

Von Neumann, memory hierarchy, IEEE 754 floating point

Step 4: Quizzes

Old computer reflection, IEEE 754 walkthrough

All steps are covered in this unit (Unit 1).

Content

- Old Computer
- Overview of Computer Organization vs. Architecture
- Basic components of a computer (**CPU**, Memory, I/O, Storage)
- 3 Different Architectures Explanation
- Floating-point representation (IEEE 754)

PART A

Computer Architectures

How hardware designers and computer architects think differently about building computers.

Computer Hardware

Hardware: What hardware do you know?

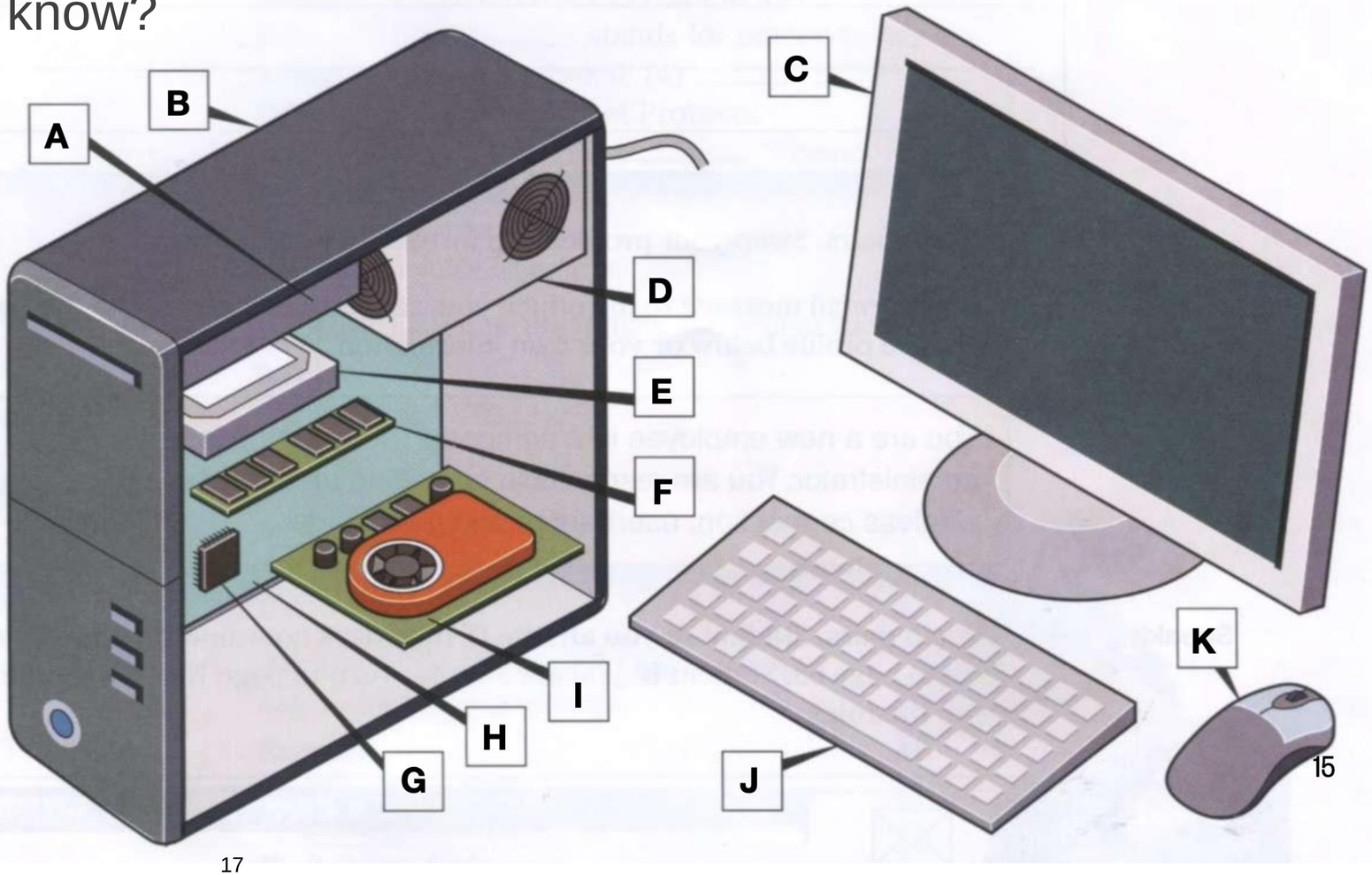
- . . .





Label the diagram with the correct items 1–11.

1	monitor C	6	expansion cards (video, graphic) I
2	case B	7	power supply unit D
3	motherboard H	8	optical disk drive A
4	CPU (Central Processing Unit or Processor) G	9	hard disk drive E
5	main memory (RAM) F	10	keyboard J
		11	mouse K



Computer Hardware

Hardware Designer

- thinks about
  - circuits,
  - components,
  - timing,
  - functionality,
  - ease of debugging



Key Points: Computer Architectures

- **Hardware designers think about circuits, components, timing, and debugging**
- **Computer architects think about high-level components and performance**
  - How components fit together to deliver performance
- **Organization (how it works) complements architecture (what it does)**

PART B

Old Computer

A look back at earlier computers (1980s-1990s) to see how far architecture has come.

Old Computer





source: https://www.youtube.com/watch?v=PF7EpEnglgk

EXERCISE

Quiz 1: Old Computer

After watching, please look at these questions:

- What did you learn from this video?
- What is the evolution of the computer?
- What are the differences between today’s computer and the old computer?
- What are today’s computers’ higher-capability pieces of equipment?

EXERCISE

Quiz 1: Old Computer

Before watching, please look at these questions and find the answers while watching the video:

- How to turn on the old computer?
- How many steps do we have to take? What are they?
- What are the procedures (indicating a sequence) to start using the old computer?
- How old is this computer?
- What pieces of equipment could not be found in the old computer?
- Can we use the computer to support doing homework?
- How to do math problems?
- How to play the game?

Old Computer

Comparing computer architectures from the 1970s-1980s to modern systems in a computer architecture course is a great way to understand how technology has evolved.

Feature	1970s-1980s	2020s
CPU Width	8-bit/16-bit	64-bit
Clock Speed	1-10 MHz	2-5 GHz
Memory	KB-MB	GB-TB
Storage	Floppy disks (KB)	SSDs (TB)
Parallelism	Single-core	Multi-core + SIMD
OS Support	Minimal (MS-DOS, CP/M)	Advanced (Linux, Windows)
Networking	None or slow (300 baud modem)	Gigabit Ethernet, 5G



Key Points: Old Computer

- **Computers evolved from room-sized machines to modern portable systems**
  - 1970s-80s machines needed manual multi-step boot procedures
- **Comparing old and new systems shows how architecture has evolved**
- **Core concepts (CPU, memory, I/O) persist across generations**

PART C

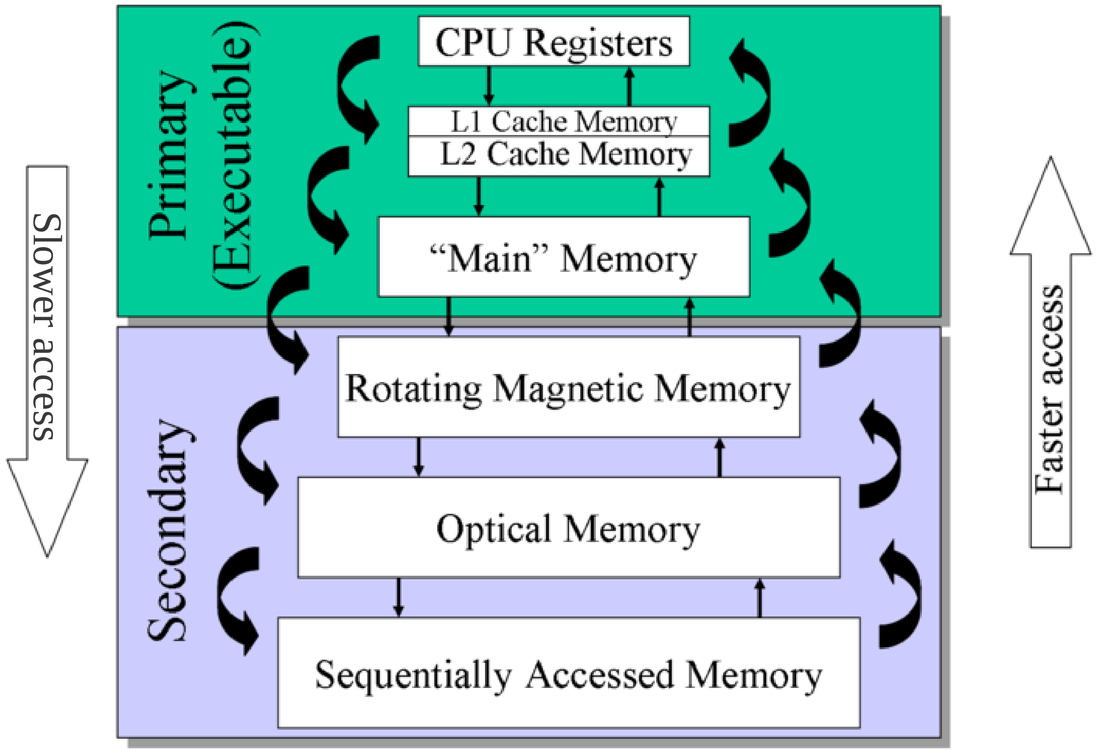
Secondary Storage Architecture

Inside HDDs, SSDs, hybrids, optical discs and tape — how data is stored long-term.

Computer Hardware

Hardware: Architecture

- Storing data in a computer

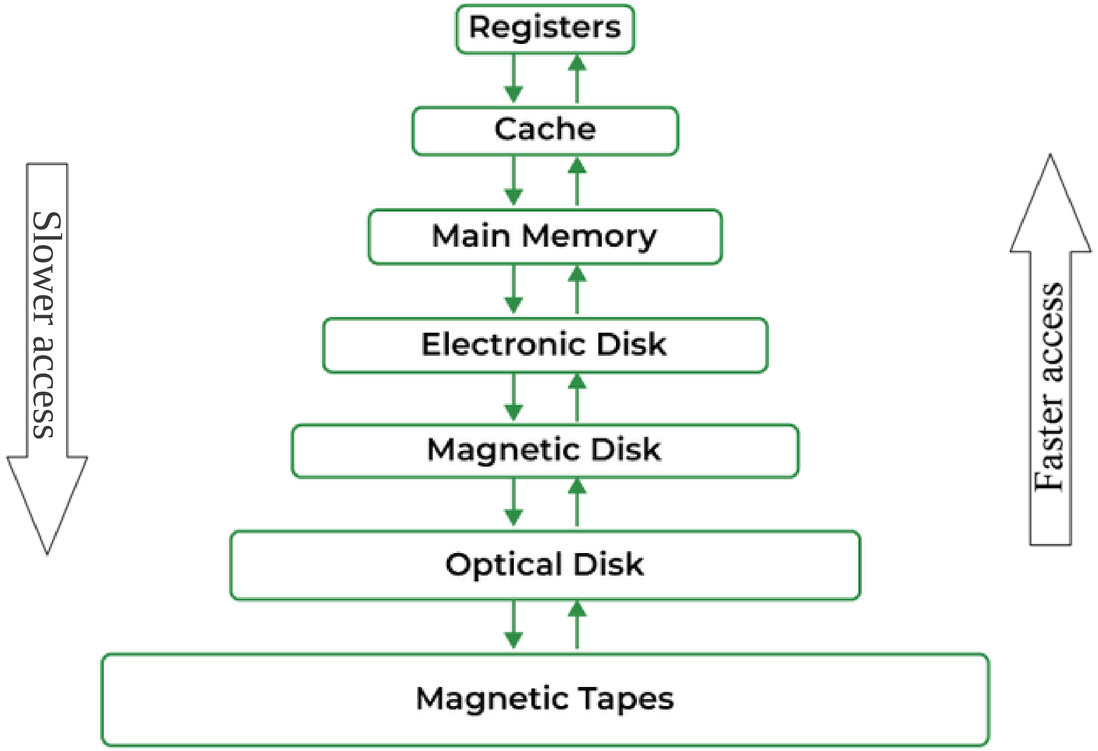


source: *https://infonw.laurensstoop.nl/Memory/mem_hw.html*

Computer Hardware

Hardware: Architecture

- Storing data in a computer



source: *https://infonw.laurensstoop.nl/Memory/mem_hw.html*

Computer Hardware

Hardware: **What are they**?

- …



inside





with cover shields

Computer Hardware

Hardware: Traditional Storage Devices

- Hard disk drive (**HDD**)



inside



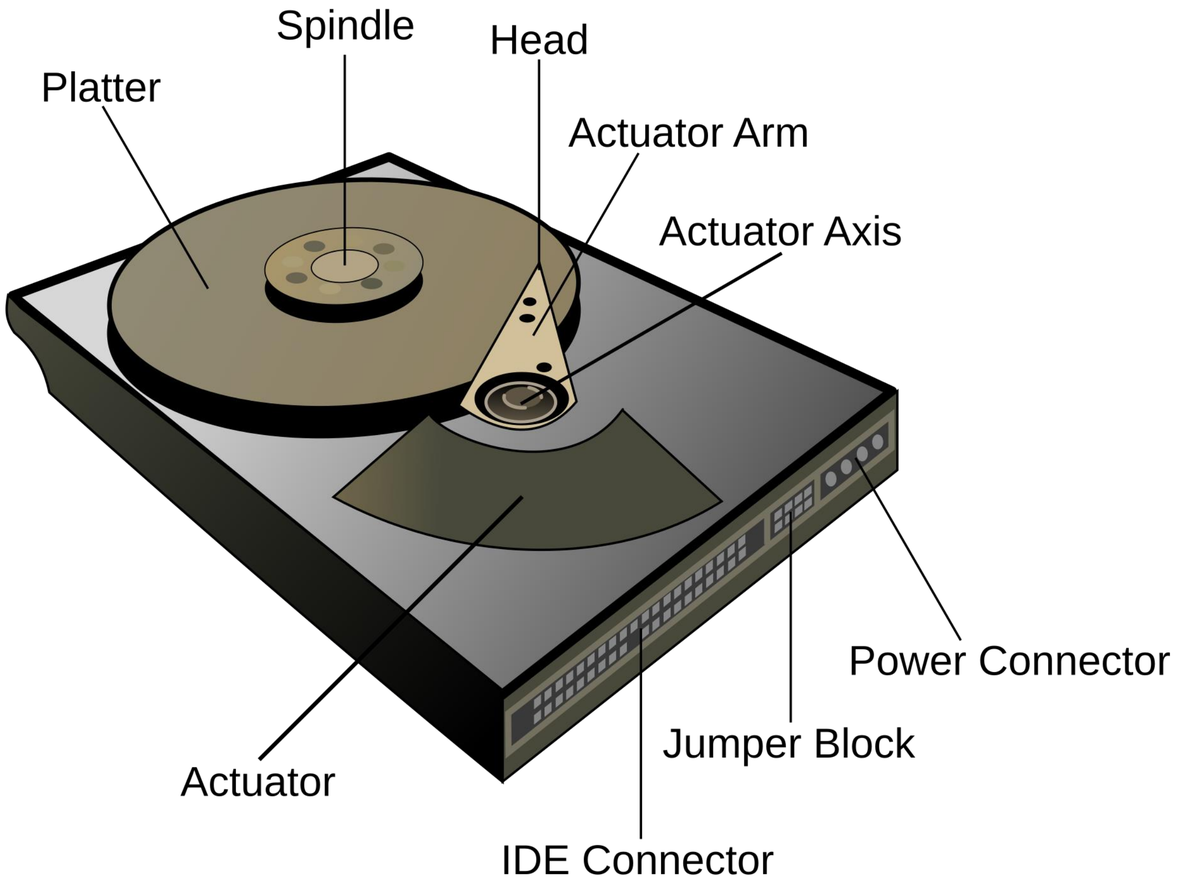
with cover shields

Computer Hardware

Hardware: Traditional Storage Devices

- In modern drives have
  - one head for each magnetic platter surface on the spindle, mounted on a common arm.

- What's happening during the read/write process?
  - An **actuator arm or access arm** moves the **heads** on an arc roughly radially across the platters as they spin.
  - **Each head can access (Read/Write)** almost the entire surface of the platter as it spins.

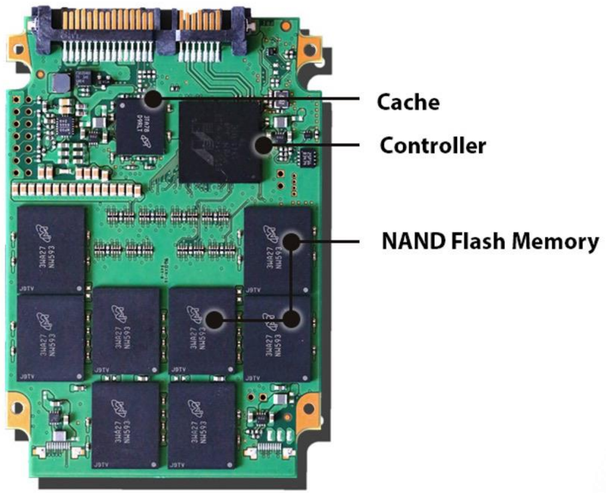


source: https://commons.wikimedia.org/wiki/File:Hard_drive.svg

Computer Hardware

Hardware: **What are they**?

- …



inside

source: https://www.backblaze.com/blog/ssd-upgrade-guide/



with cover shields



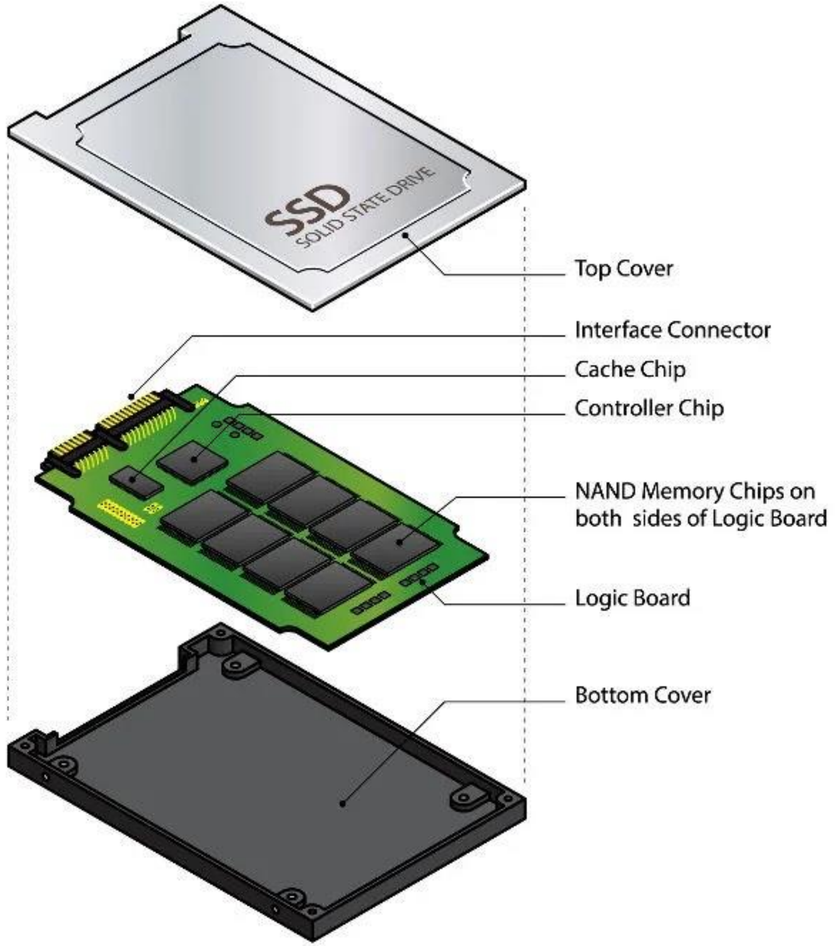
Computer Hardware

Hardware: Solid State Drives (SSDs)

- Solid State Drives (SSDs) are storage devices that contain **non-volatile flash memory**.
- Solid state drives are superior to mechanical hard disk drives (HDDs) in terms of **performance, power use, and availability**.



with cover shields



inside

non-volatile (adj) - not likely to change suddenly and unexpectedly.

source: https://www.researchgate.net/figure/Deconstruction-of-a-generic-solid-state-drive-SSD-showing-its-usage-of-flash-memory_fig10_364516996

Computer Hardware

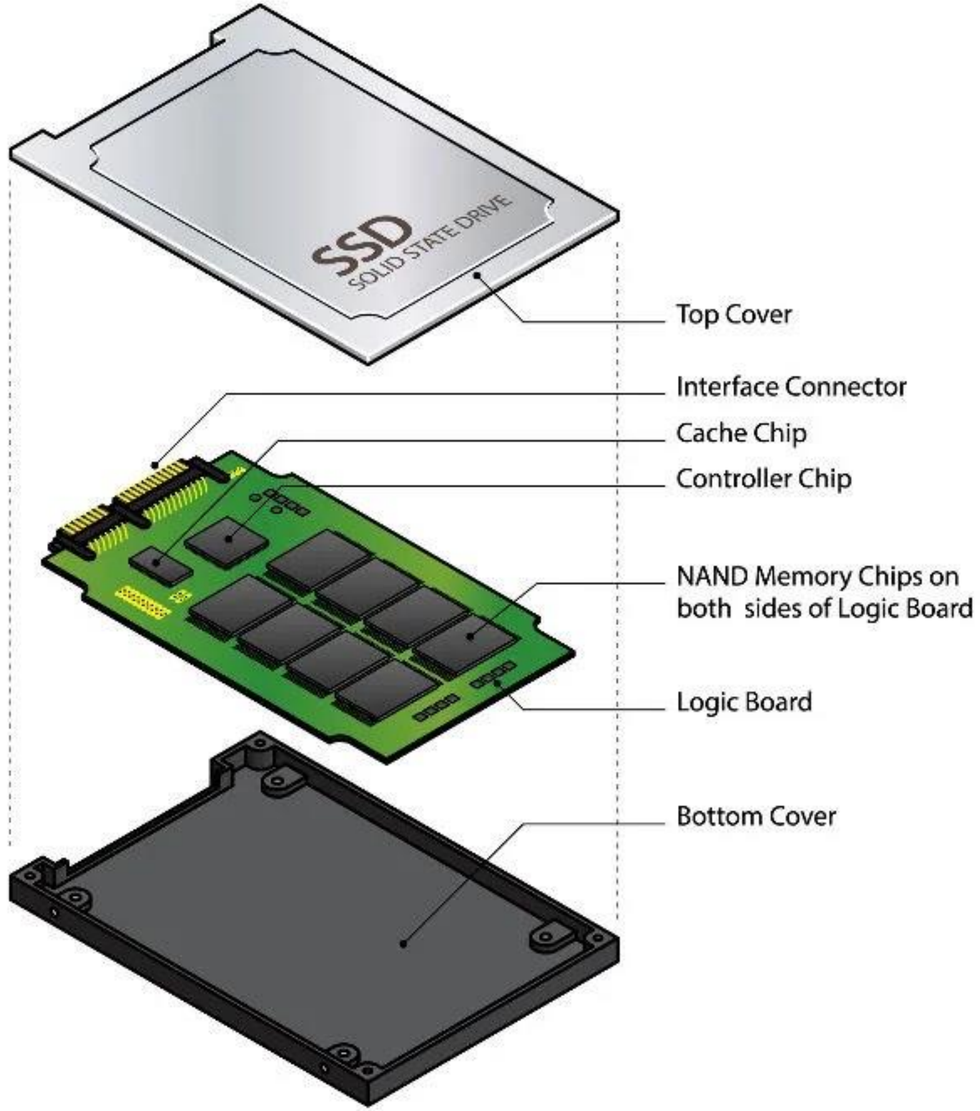
Hardware: Solid State Drives (SSDs)

- SSDs **consume less power** compared to hard disk drives.
- Because SSDs **do not have moving parts**, they generate **less heat** compared to HDDs.

- It has "**mass storage"** in an **array of non-volatile memory chips**.

- They **retain** their contents when powered off.
- The chips are commonly called **Flash memory**.

retain (v) - to keep or continue to have something.



inside

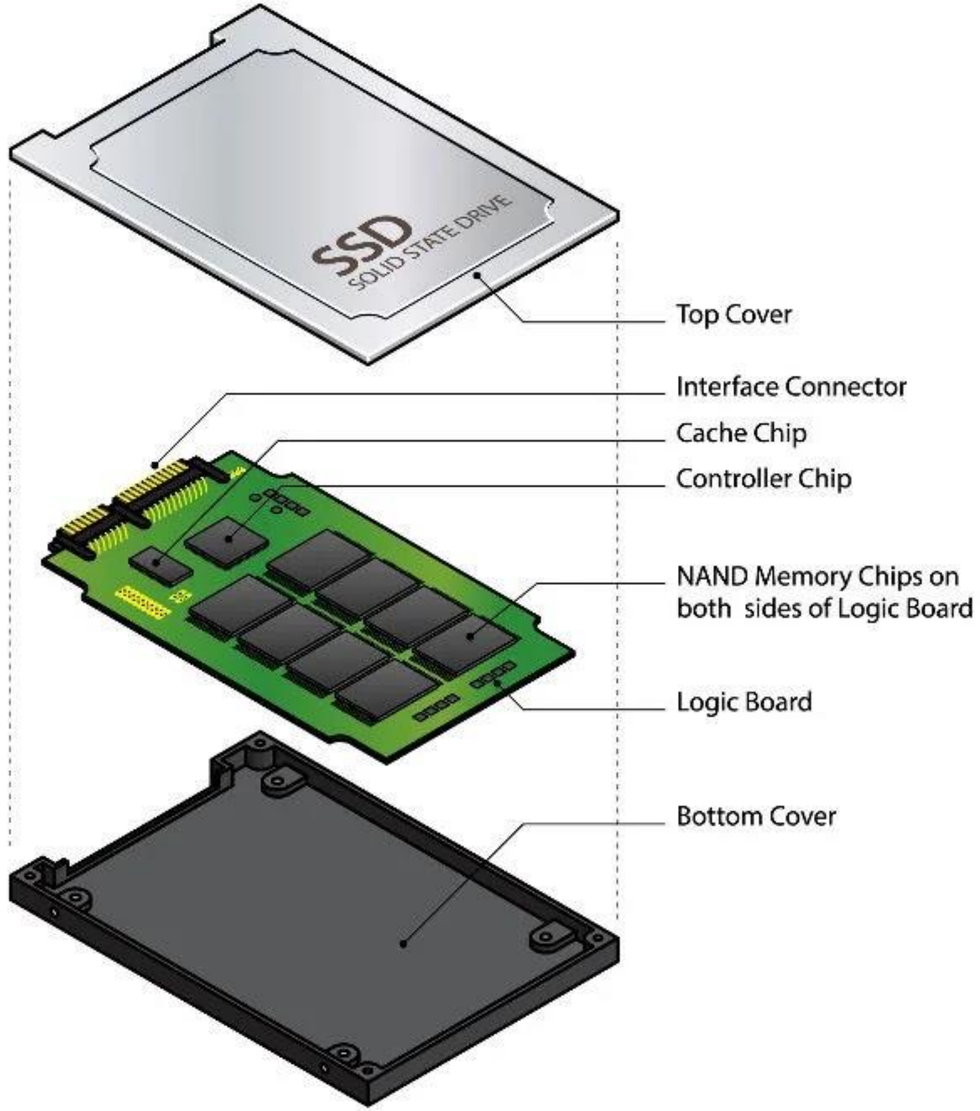
source: https://www.researchgate.net/figure/Deconstruction-of-a-generic-solid-state-drive-SSD-showing-its-usage-of-flash-memory_fig10_364516996

Computer Hardware

Hardware: Solid State Drives (SSDs)

- The **number and capacity** of the individual chips vary directly in relationship to the **SSD**'s capacity.

- The larger the capacity of the **SSD**,
  - The larger the **capacity and** the **greater the number** of Flash memory chips.



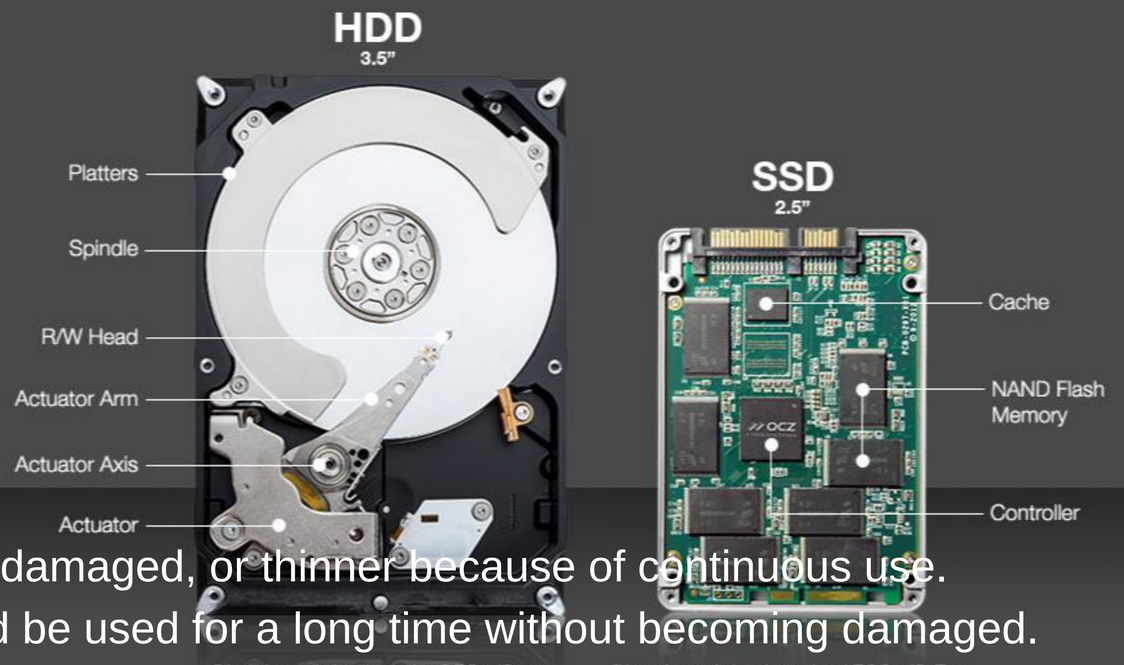
inside

source: https://www.researchgate.net/figure/Deconstruction-of-a-generic-solid-state-drive-SSD-showing-its-usage-of-flash-memory_fig10_364516996

Computer Hardware

Hardware: Components Comparison

- Mechanical vs. Electronic:
  - **HDDs** are electromechanical, with moving parts → slower, prone to **wear**.
  - **SSDs** are fully electronic, with no moving parts → faster, more **durable**.



source: https://ictbestet.click/product_details/474460.html

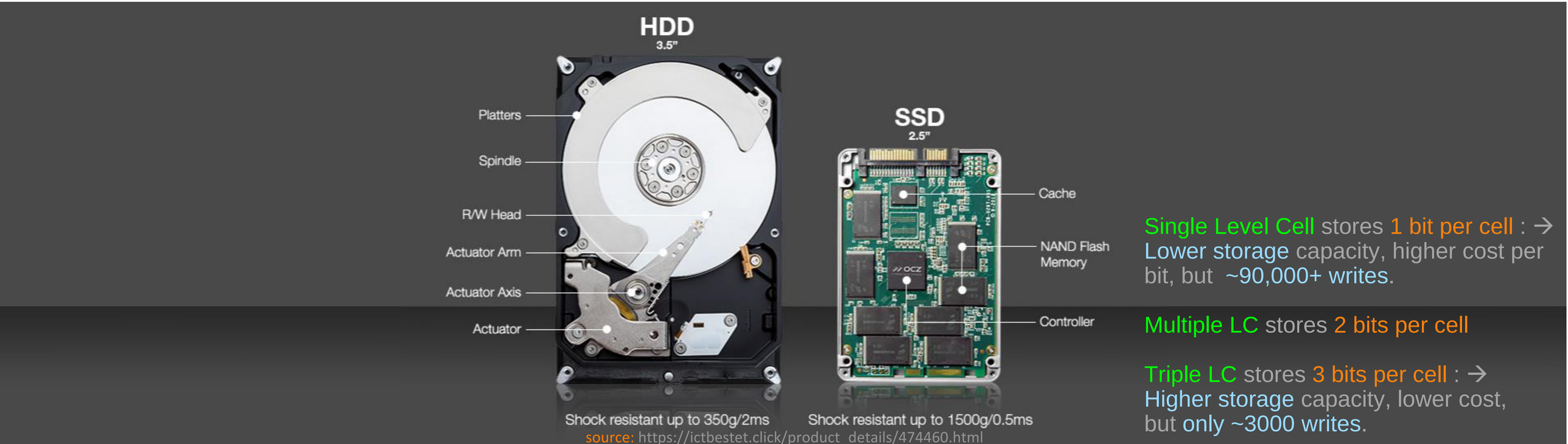
wear (v) - to become weaker, damaged, or thinner because of continuous use.

durable (adj) - able to last and be used for a long time without becoming damaged.

Computer Hardware

Hardware: Components Comparison

- Performance:
  - **HDD**: Speed depends on spindle RPM and head movement.
  - **SSD**: Speed depends on **NAND Flash Memory** type (SLC/ MLC/ TLC) and controller.



Single Level Cell stores 1 bit per cell : → Lower storage capacity, higher cost per bit, but ~90,000+ writes.

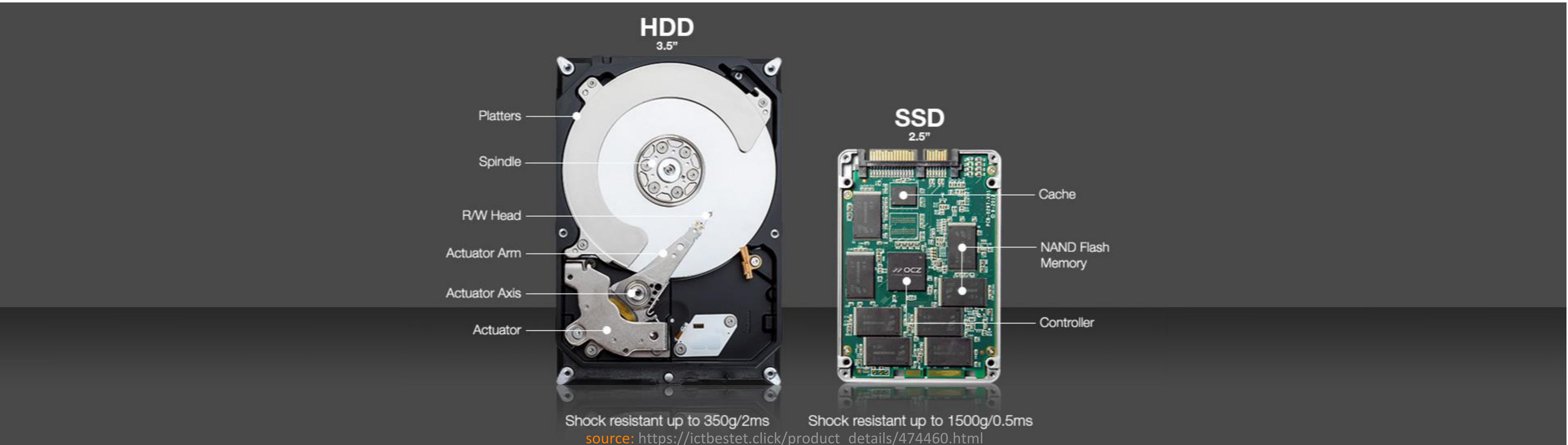
Multiple LC stores 2 bits per cell

Triple LC stores 3 bits per cell : → Higher storage capacity, lower cost, but only ~3000 writes.

Computer Hardware

Hardware: Components Comparison

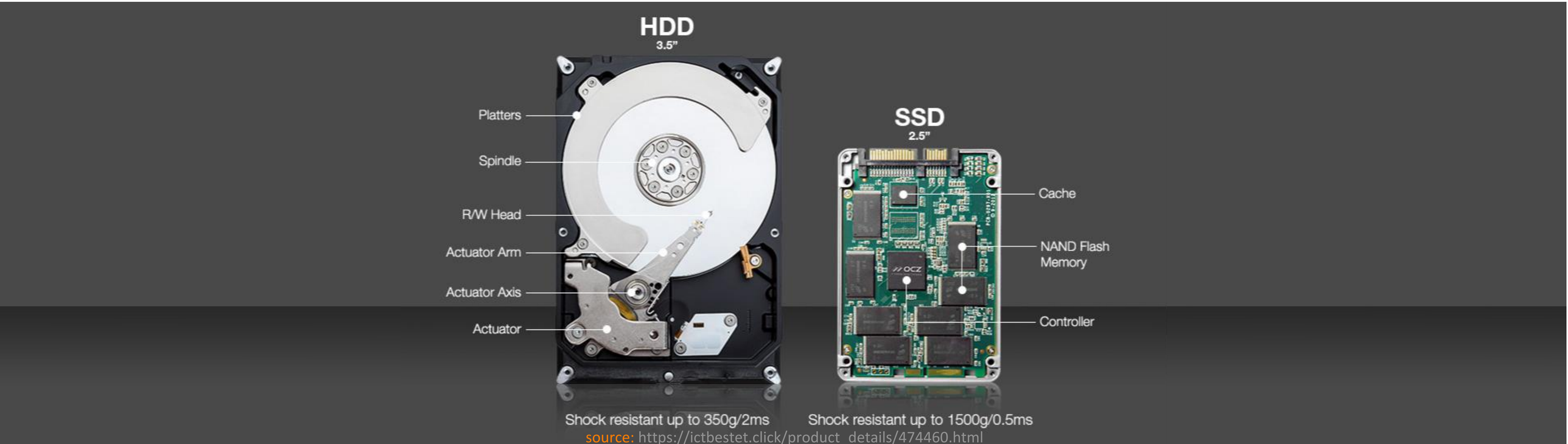
- Components Unique to **HDD**:
  - **Platters, spindle, actuator arm, and R/W head** are **absent** in **SSDs**.



Computer Hardware

Hardware: Components Comparison

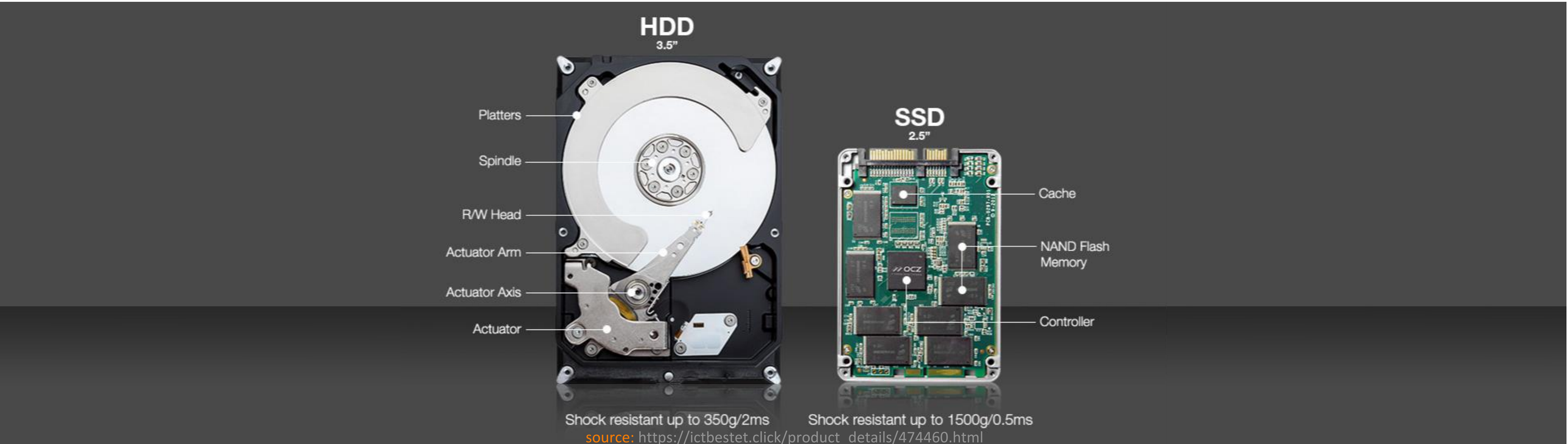
- Components Unique to **SSD**:
  - **NAND flash** and **advanced memory management controllers** (with wear-leveling) are absent in HDDs.



Computer Hardware

Hardware: Components Comparison

- Shared Features:
  - Both use **cache** (though **SSD cache is typically larger**).
  - Both require controllers, but **SSD controllers are more complex**.



Computer Hardware

Hardware: Components Comparison

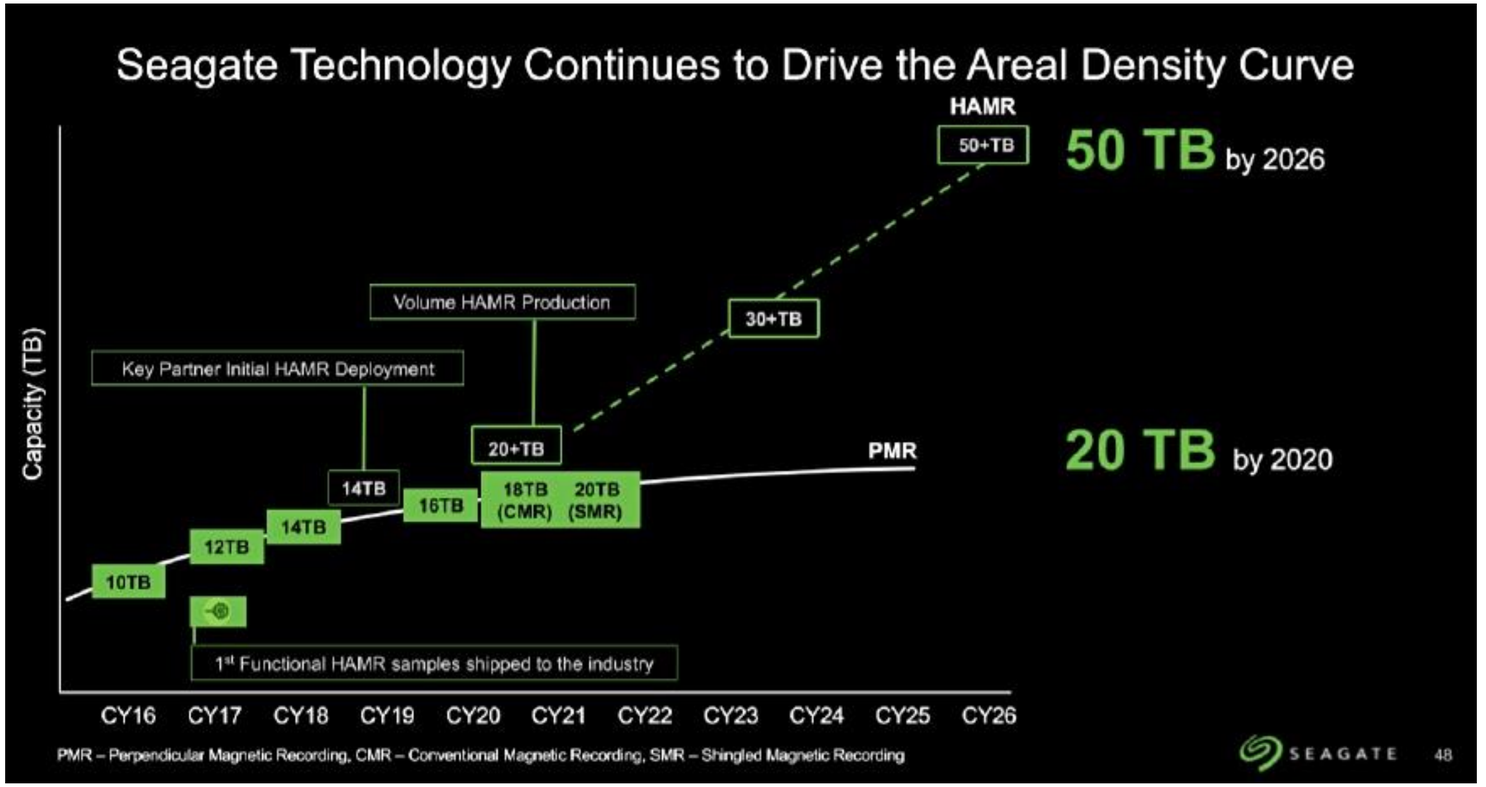
Component	HDD (Mechanical)	SSD (Electronic)	Similarities/Differences
Storage Medium	Platters (magnetic disks)	NAND Flash Memory (chips)	Difference: HDD uses spinning magnetic platters; SSD uses silicon-based memory cells.
Read/Write Head	R/W Head (moves over platters)	Controller (manages data flow)	Difference: HDD has physical heads; SSD has no moving parts—data is accessed electronically.
Actuator Arm	Moves R/W head across platters	N/A	Difference: Unique to HDDs for mechanical movement.
Spindle	Rotates platters at high speed	N/A	Difference: HDDs rely on spindle speed (e.g., 5400/7200 RPM); SSDs have no spinning parts.
Actuator	Controls arm movement	N/A	Difference: HDD uses actuators for positioning; SSDs require no mechanical motion.
Cache	Small DRAM buffer (e.g., 64MB)	DRAM Cache (e.g., 1GB)	Similarity: Both use cache to speed up frequent data access. Difference: SSD cache is larger.
Controller	Manages I/O and error correction	Controller (more advanced)	Similarity: Both have controllers. Difference: SSD controllers handle wear-leveling, encryption, etc.

Computer Hardware

Hardware: **Are HDDs obsolete?**

- With Seagate's **Heat-Assisted Magnetic Recording (HAMR)** technology, the number of bits that can be stored on a disk's surface **is expected to continue to grow exponentially in the future**.

- Seagate have 32TB HDDs are currently available in the market **in 2026**, and up to 50TB by Late 2027.

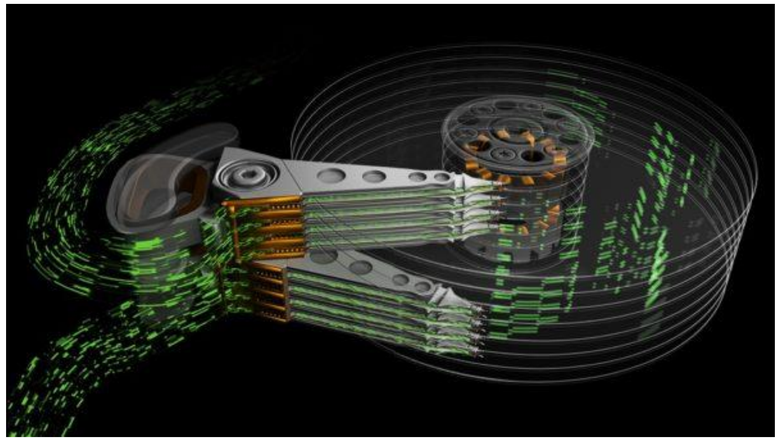


source: Seagate

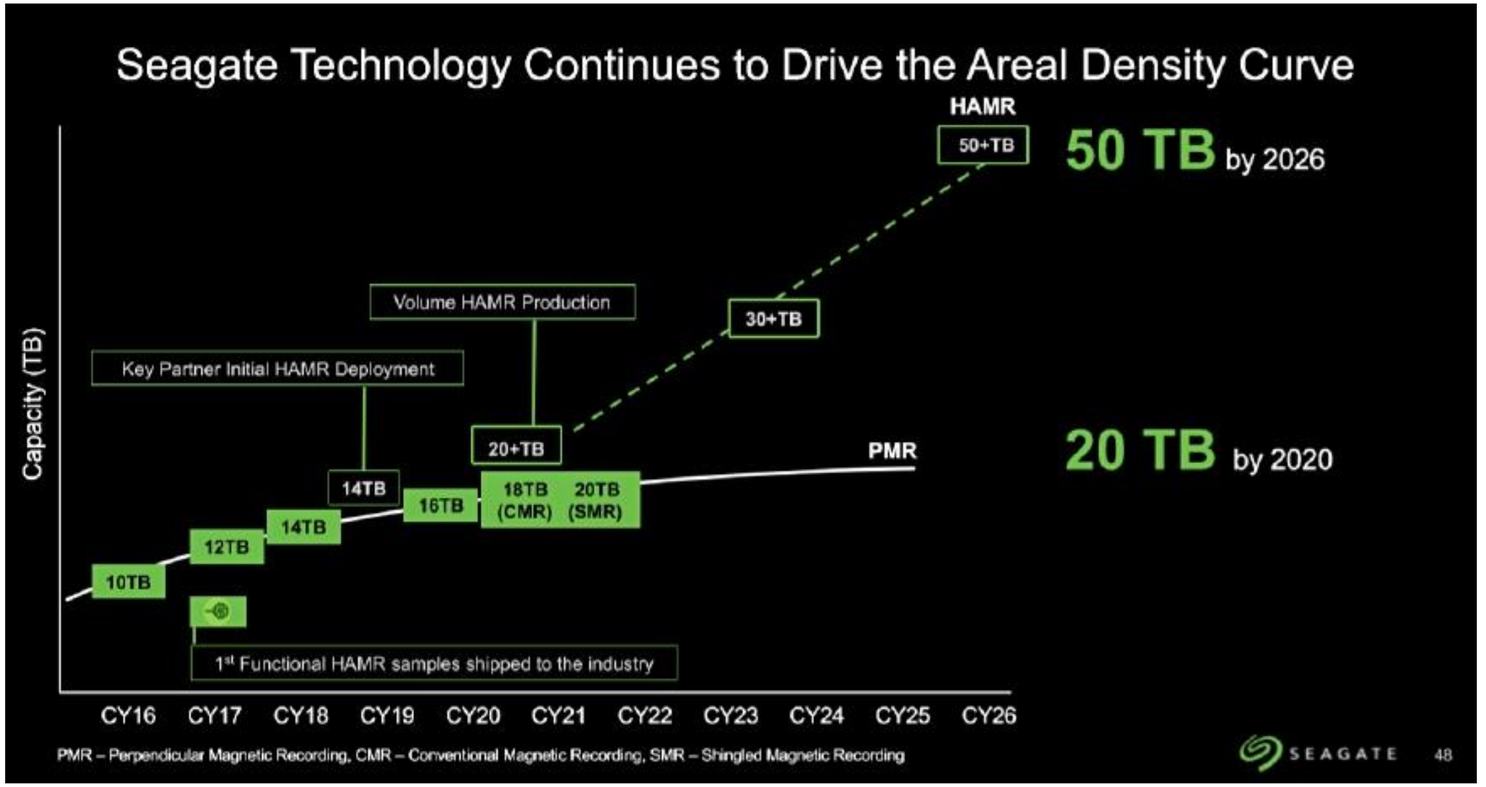
Computer Hardware

Hardware: Are HDDs obsolete?

- New technologies like Multi-Actuator Technology will also enable faster access to data on HDDs.



- This trend of rapid capacity growth on HDDs is expected to continue until 2034.

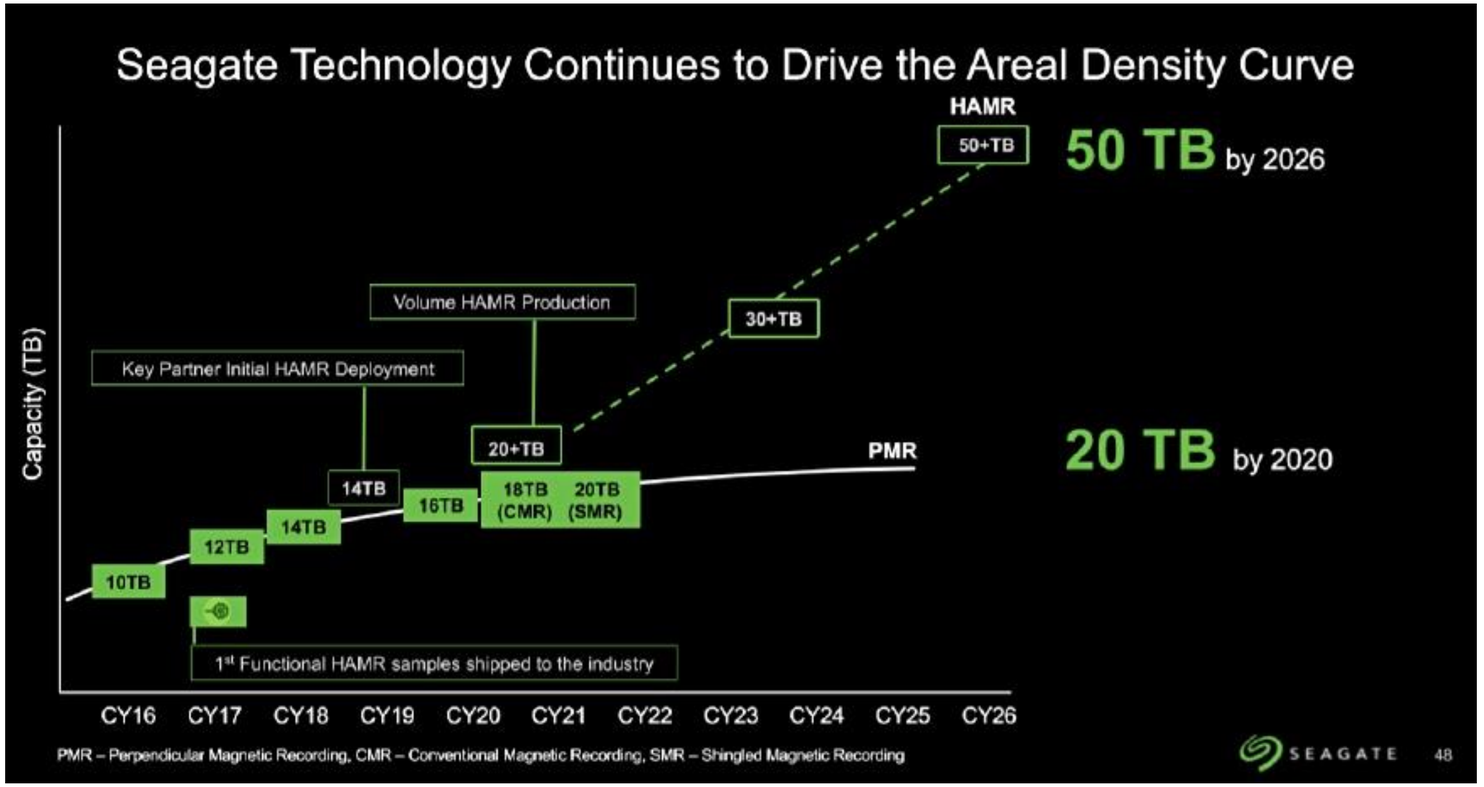


source: Seagate

Computer Hardware

Hardware: Are HDDs obsolete?

- Seagate believes that the rapidly increasing capacities of HDDs will continue to push SSDs using newer technologies like QLC Flash (4 bits/cell) or **3D NAND** with increasingly higher layers.

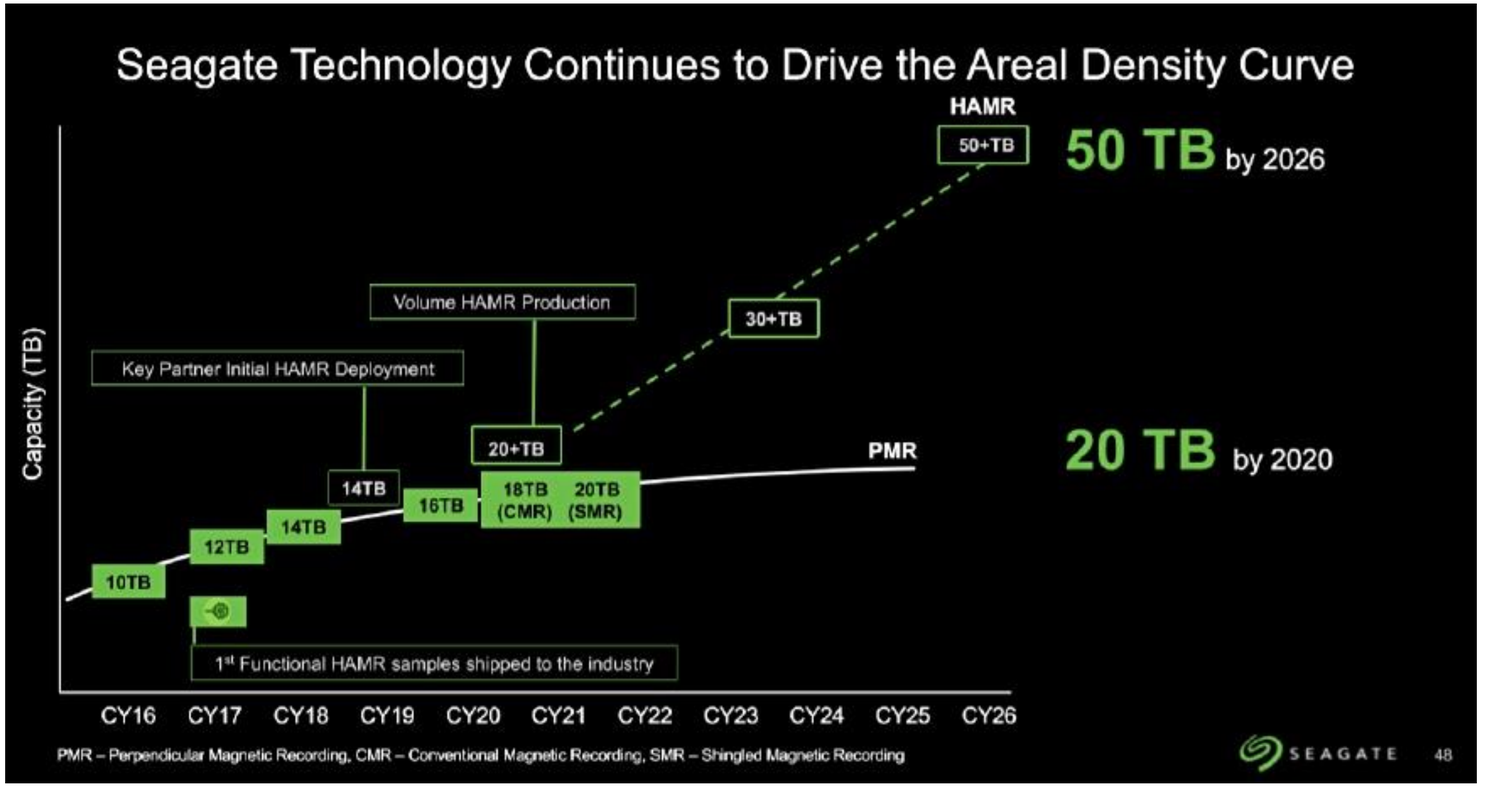


source: Seagate

Computer Hardware

Hardware: Are HDDs obsolete?

- And while some analysts believe that once the price-to-capacity ratio of SSDs is less than five times that of HDDs, many users will move to SSDs.
- Seagate remains confident that HDDs will remain essential for the market in the future.



source: Seagate

Computer Hardware

Hardware: **What are they?**

- …

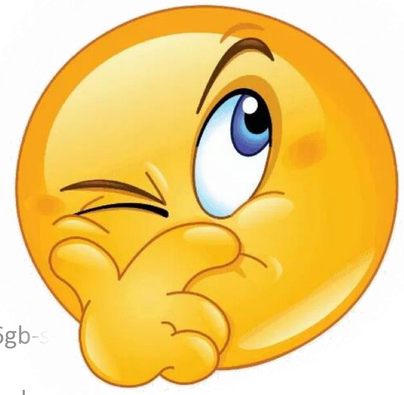




inside

source: https://drivesolutions.com/w-d-black-sshd-1tb-5400rpm-16gb-nand-flash-sata-6gb-s-16mb-cache-2-5-7mm-solid-state-hybrid-drive-wd10s12x.html?srsltid=AfmBOorp0stg_XJpXuzv_fqz0f1J8jer303BVaCYJ7xfopaosAAIujml

source: https://recoverit.wondershare.com/hard-drive/what-is-hybrid-drive.html







with cover shields

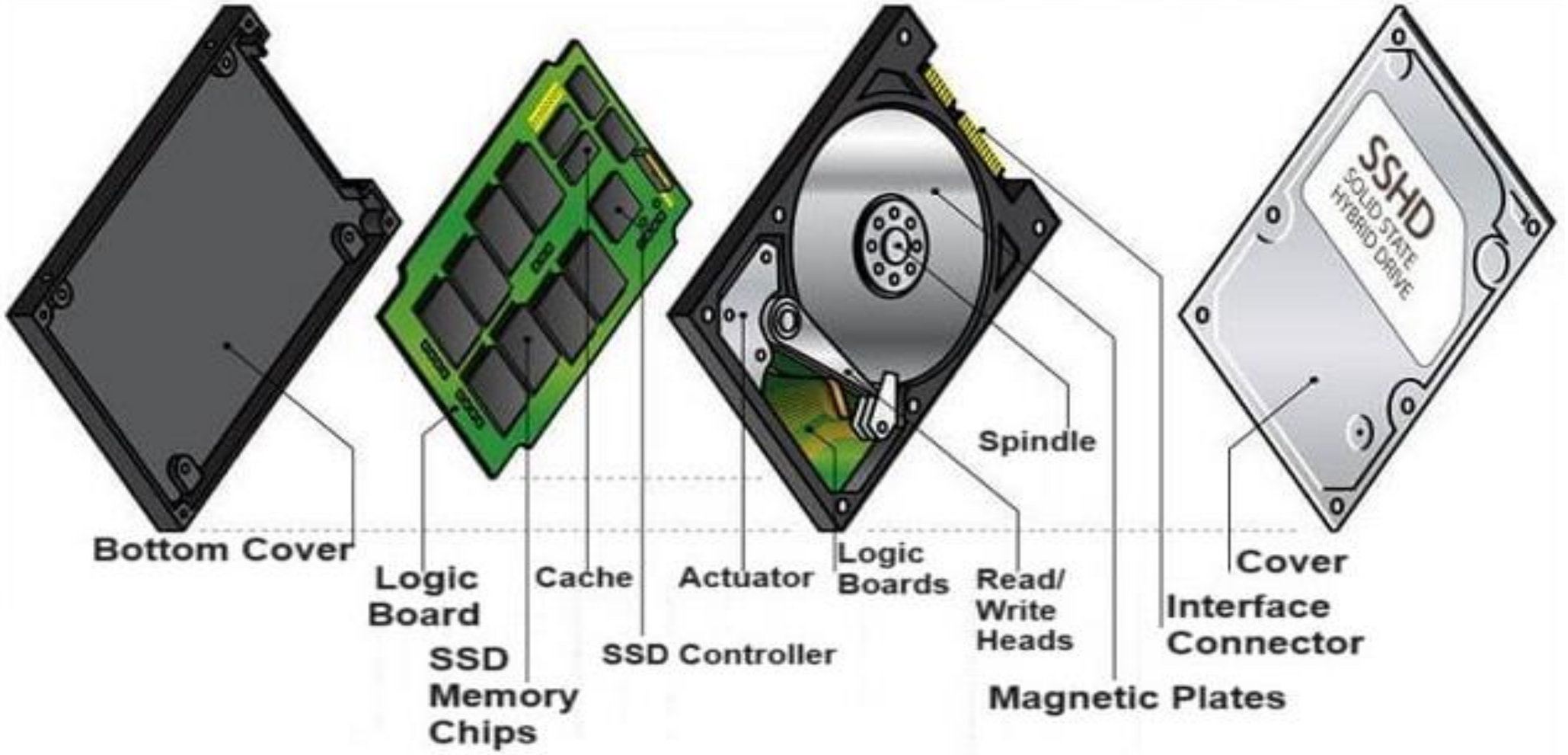
rce: https://drivesolutions.com/w-d-black-sshd-1tb-5400rpm-16gb-nand-flash-sata-6gb-s-16mb-cache-2-5-7mm-solid-state-hybrid-drive-wd10s12x.html?srsltid=AfmBOorp0stg_XJpXuzv_fqz0f1J8jer303BVaCYJ7xfopaosAAIujml

Computer Hardware

Hardware: **What is it?**

- …



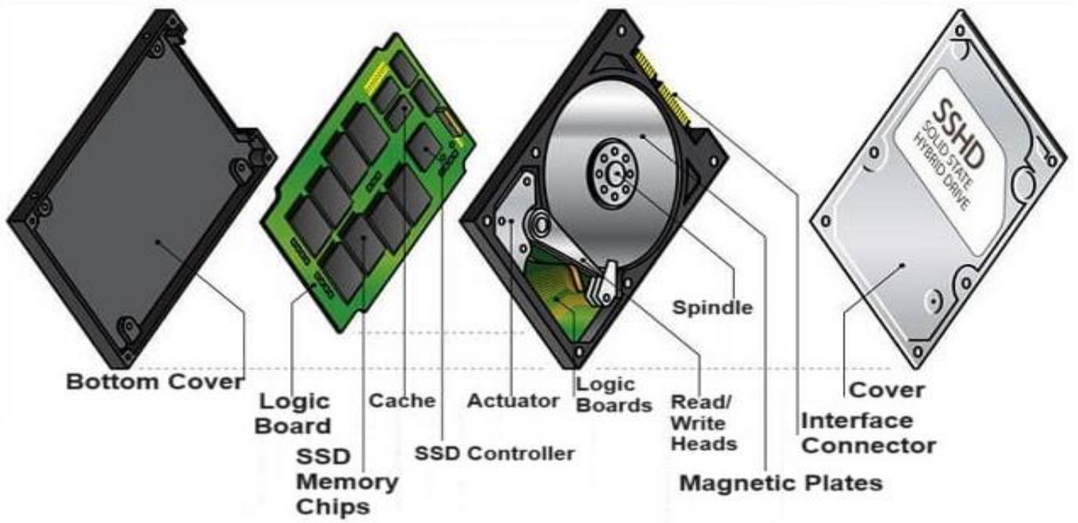


source: https://electricalfundablog.com/hhd-sshd-solid-state-hybrid-drive/

Computer Hardware

Hardware: **Hybrid Hard Drives (HHDs)**

- A Hybrid Hard Drive (HHD) is a type of computer hard drive that **combines** the features of a traditional Hard Disk Drive (**HDD**) and a solid-state drive (**SSD**).
- It is also known as **Solid State Hybrid Drive (SSHD).**
- They are so popular because they fulfill both the requirements of speed and capacity of a system.
- It typically has a larger storage capacity, like an **HDD**, and faster data access speeds, like an **SSD**.
- This combination aims to provide a balance between storage capacity and performance.

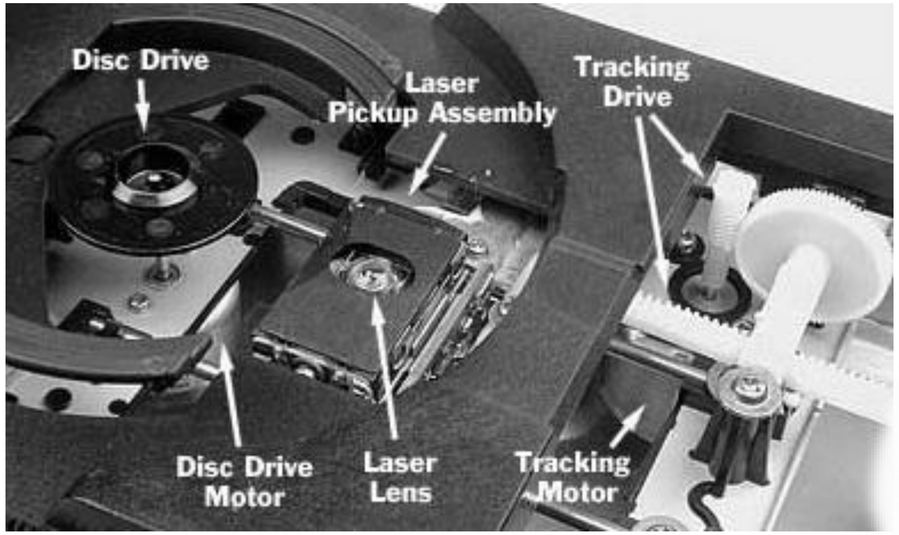


source: https://electricalfundablog.com/hhd-sshd-solid-state-hybrid-drive/

Computer Hardware

Hardware: **What are they**?

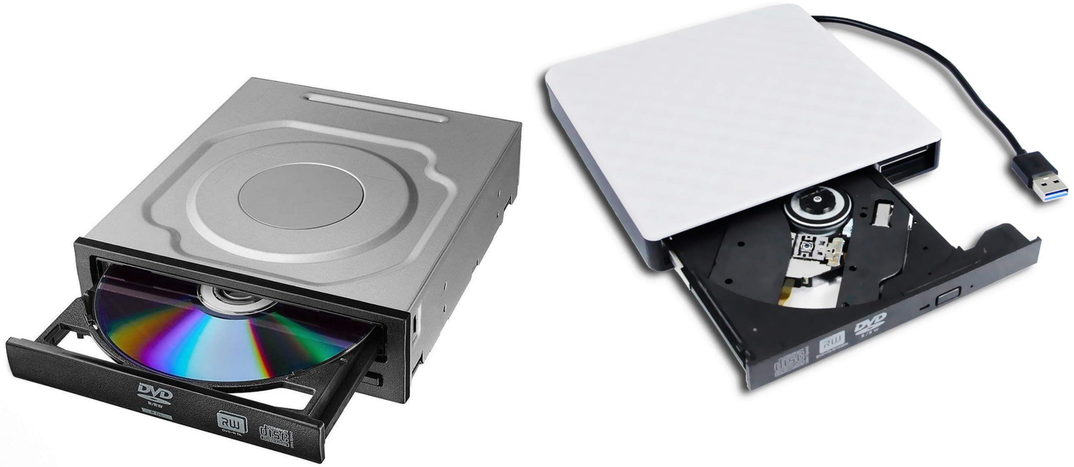
- …



inside

source: https://www.researchgate.net/figure/Optical-Disc-reading-device_fig4_37181316





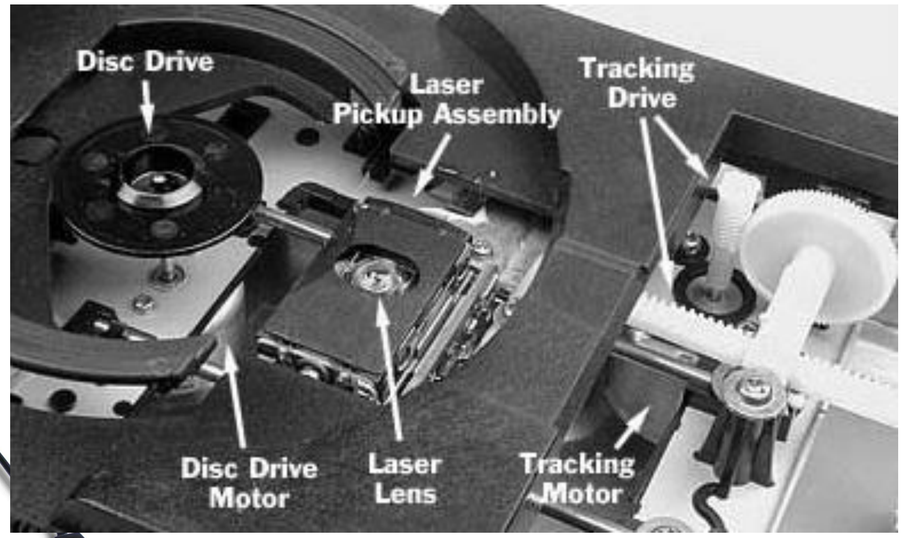
with cover shields

Computer Hardware

Hardware: Optical Disc Drives (ODDs) and Optical Discs

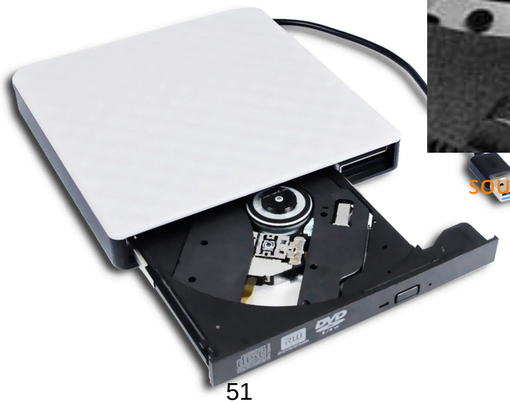
- An **optical disc drive (ODD)** is a disc drive that uses **laser light** or **electromagnetic waves** within or **near the visible light spectrum** as part of the process of reading or writing data to or from optical discs.

- Some drives can only read from certain discs.
- Other drives can both read and record
- Those drives are called **burners** or **writers** since they physically burn the data onto the discs.



source: https://www.researchgate.net/figure/Optical-Disc-reading-device_fig4_37181316

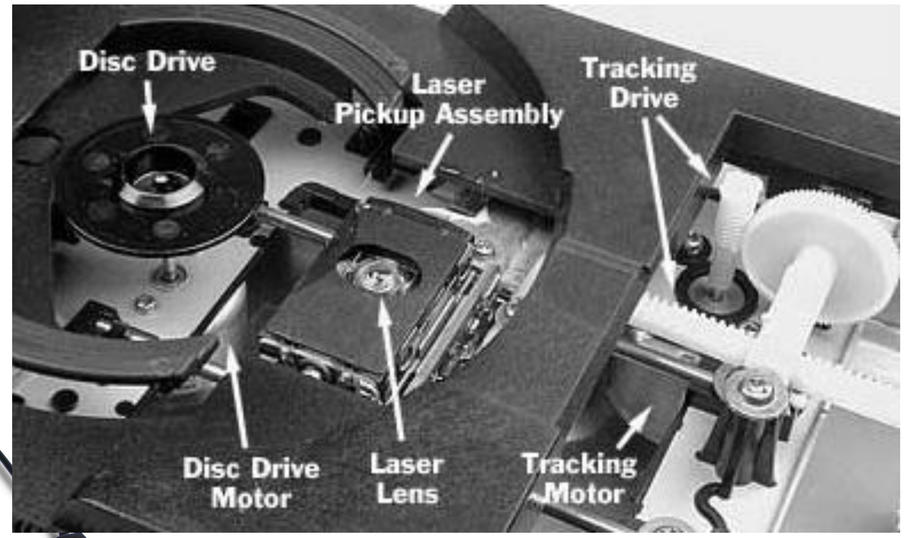




Computer Hardware

Hardware: Optical Disc Drives (ODDs) and **Optical Discs**

- Common types of optical media that can be **read and recorded** by such drives.
  - **Compact discs (CDs),**
  - Digital video discs or **digital versatile discs (DVDs)**, and
  - **Blu-ray discs**.



source: https://www.researchgate.net/figure/Optical-Disc-reading-device_fig4_37181316



Computer Hardware

Hardware: **What is this**?

- …





with cover shields

Computer Hardware

Hardware: Magnetic Tapes

- A **magnetic tape is a medium for magnetic storage** made of a **thin, magnetizable coating** on a long, narrow strip of plastic film.

- Information on magnetic tapes is often recorded in tracks.
  - The tracks are narrow, **long sequential areas** of information recorded magnetically onto the tape.
- These tracks are separate from each other and often **spaced apart from adjacent tracks**.

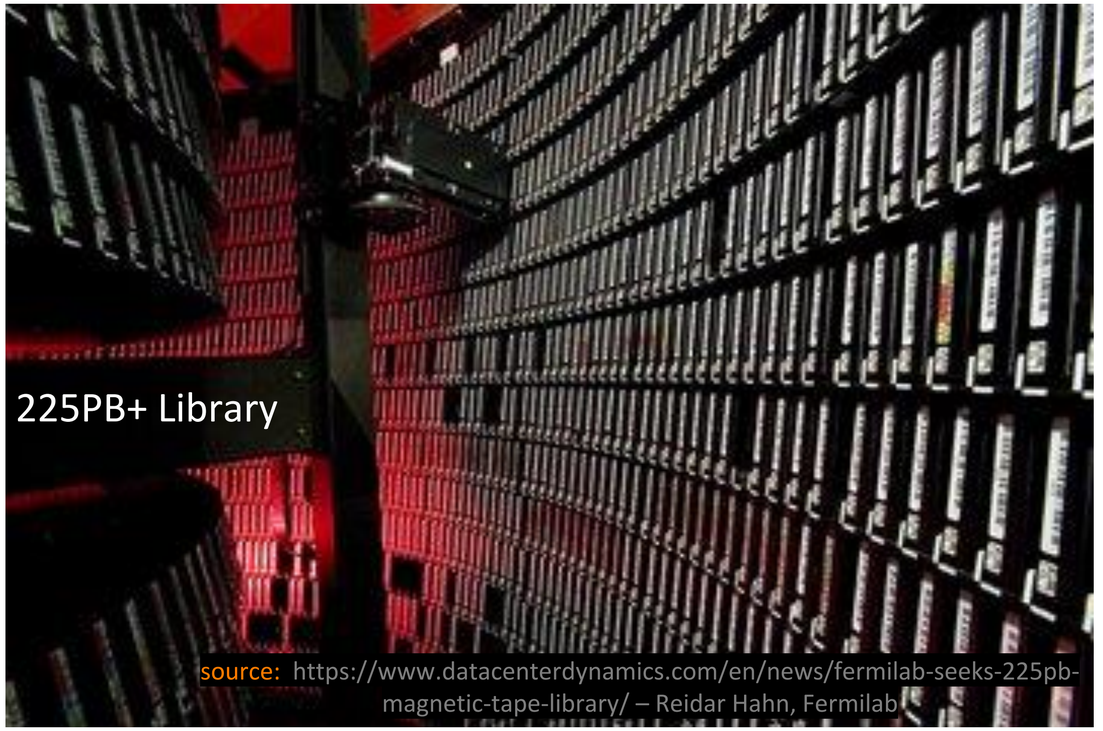


source: http://erdhritiman-mainframes.blogspot.com/2008/03/history-of-mainframe-computers.html

Computer Hardware

Hardware: Magnetic Tapes

- **Modern data centers** widely use magnetic tapes for automated backups for *disaster recovery / ransomware*,
  - mainly because they are:
    - Exceptionally cheap per terabyte
    - Require no power when idle (to store / retain data).
- Backup storage of +225 *petabytes (PB)* of data as becomes feasible and **can be retrieved** (*if slightly slowly*).



source: https://www.datacenterdynamics.com/en/news/fermilab-seeks-225pb-magnetic-tape-library/ – Reidar Hahn, Fermilab



source: https://www.tape-storage.net/en/about_tape_storage/



Key Points: Secondary Storage Architecture

- **HDDs are electromechanical: platters, spindle, actuator arm, R/W heads**
  - Speed depends on spindle RPM and head movement
- **SSDs are fully electronic, using NAND flash and advanced controllers**
  - No moving parts: faster, cooler, more power-efficient
- **Hybrid drives (HHDs) combine HDD capacity with SSD-like caching**
- **Optical discs and magnetic tape serve archival and distribution roles**

PART D

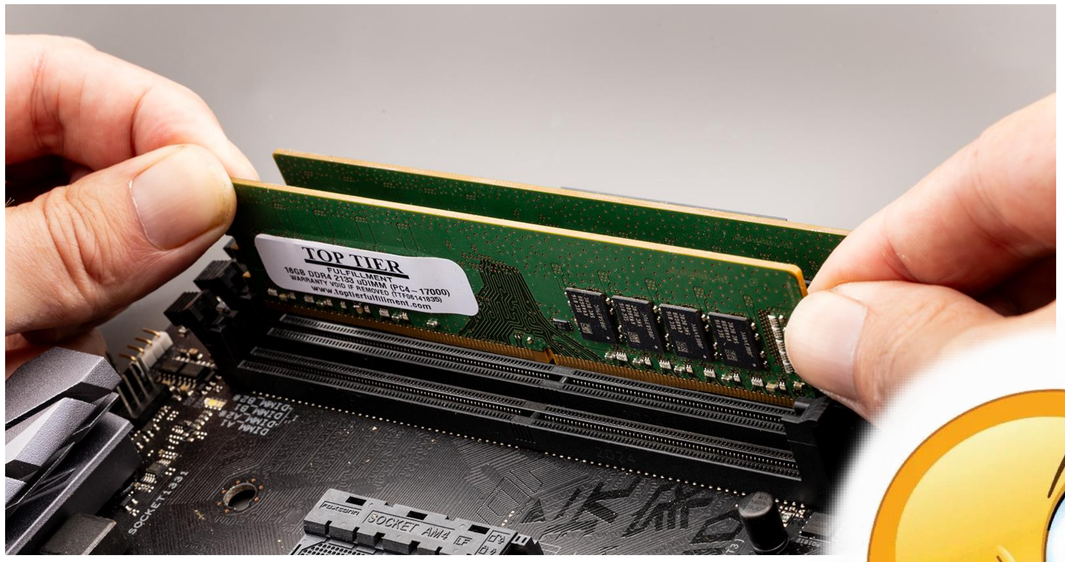
Primary Storage Architecture

RAM: the fast, short-term memory your processor works from.

Computer Hardware

Hardware: What are they?

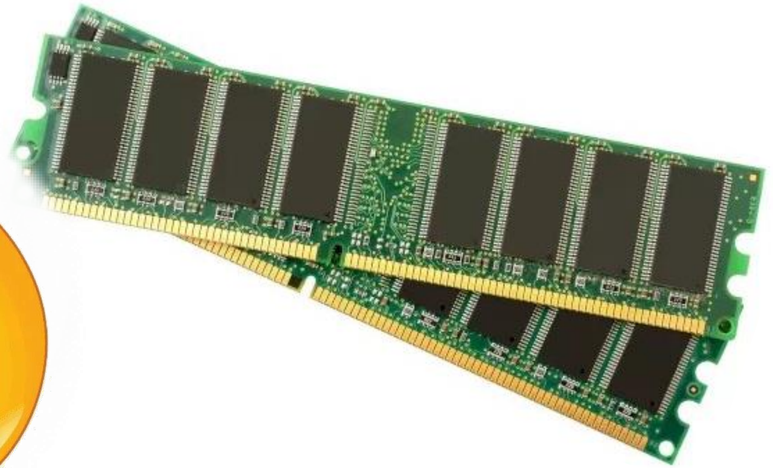
- …



source: https://biznis.ba/tech/predstavljen-najbrzi-ram-za-telefone/



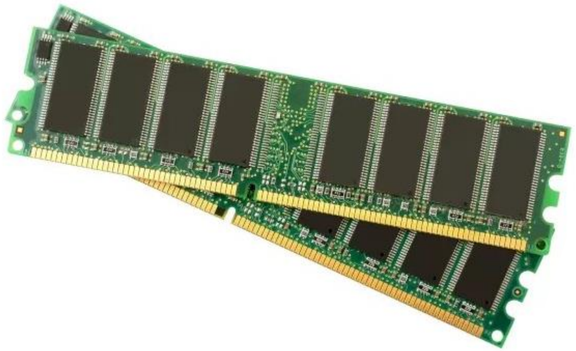




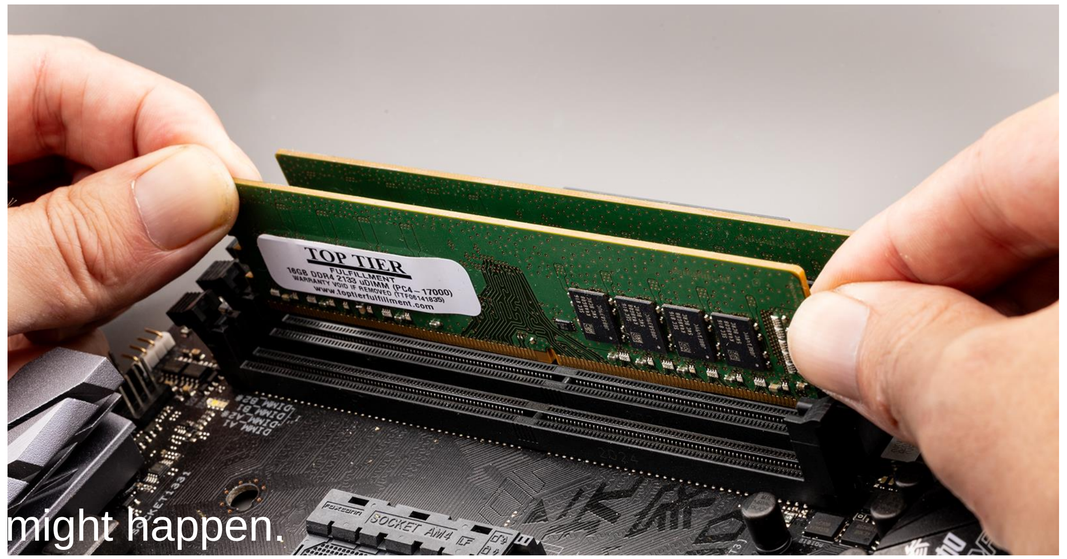
Computer Hardware

Hardware: Random Access Memory (**RAM**)

- **RAM** (random access memory) is a computer's short-term memory, where the data that the processor is currently using is stored.
- Computers can access **RAM** much faster than data on a hard disk, **SSD**, or other long-term storage devices, which is why RAM capacity is critical for system performance.







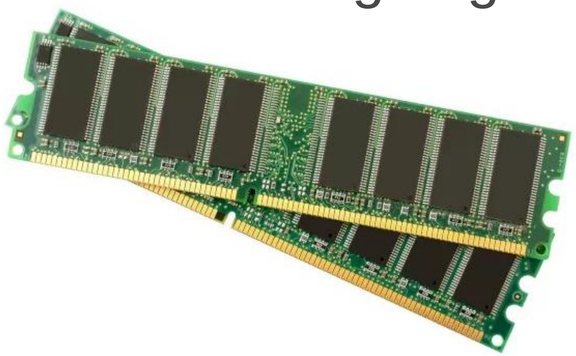
critical (adj) - of the greatest importance to the way things might happen.

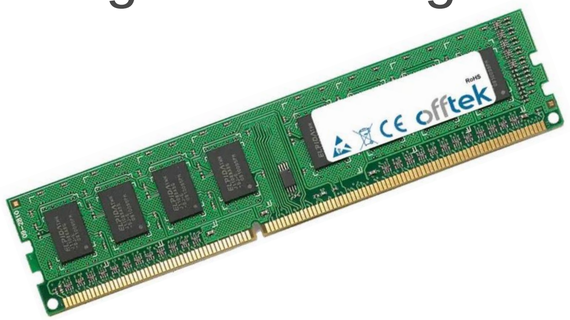
source: https://biznis.ba/tech/predstavljen-najbrzi-ram-za-telefone/86479

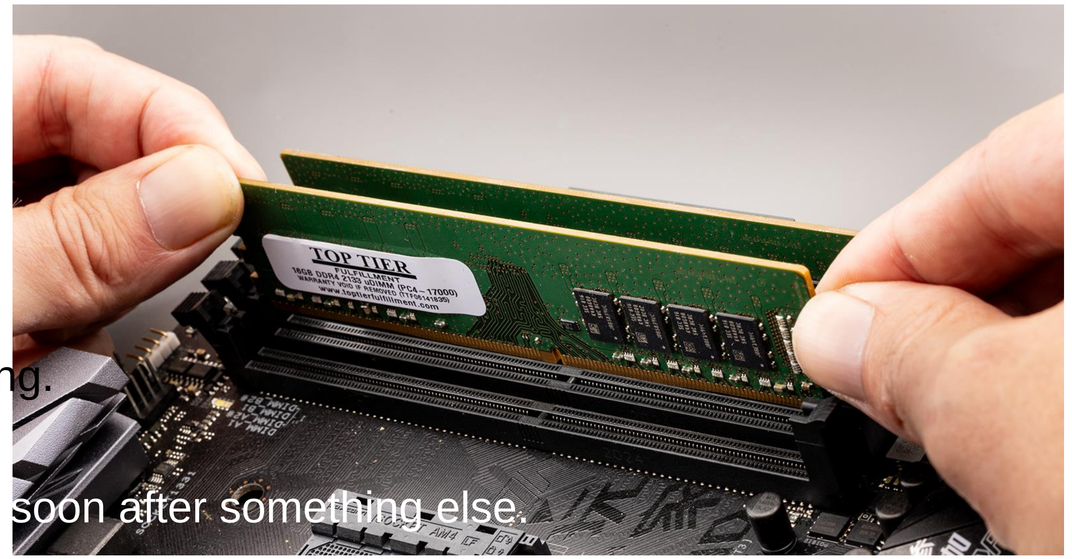
Computer Hardware

Hardware: Random Access Memory (**RAM**)

- **RAM** is a temporary memory bank where your computer stores data it needs to retrieve quickly.
- **RAM** keeps data easily accessible so your processor can quickly find it without having to go into long-term storage to complete immediate processing tasks.







source: https://biznis.ba/tech/predstavljen-najbrzi-ram-za-telefone/86479

temporary (adj) - (a r needed for a very long.

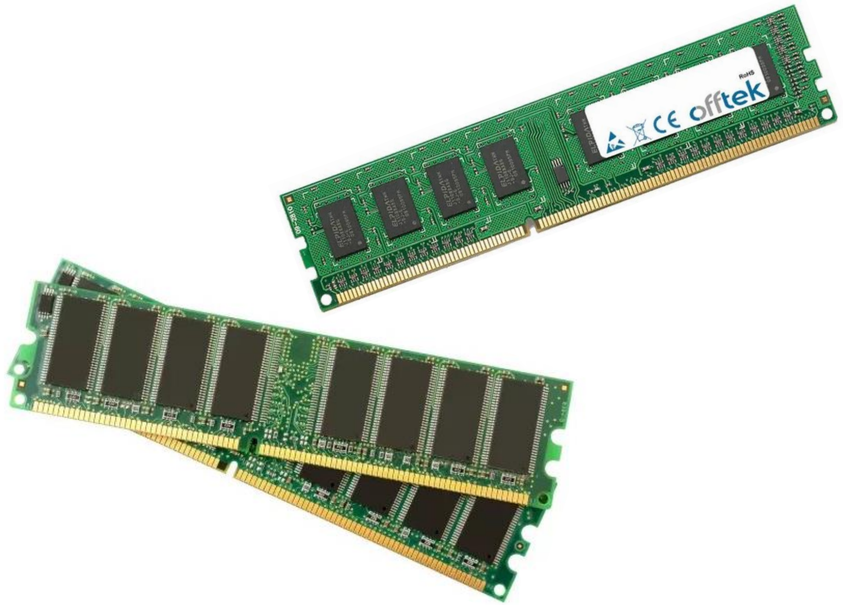
accessible (adj) - able to b ed or easily obtained.

immediate (adj) - happening or done without delay or very soon after something else.

Computer Hardware

Hardware: Random Access Memory (**RAM**)

- Every computing device has **RAM**, whether it's
  - desktop computer (running Windows, MacOS, or Linux),
  - tablet (iPad) or
  - smartphone (running Android or iOS), or even
  - IoT computing device (like a smart TV).





Key Points: Primary Storage Architecture

- **RAM is the computer's short-term memory for data in active use**
  - Volatile: contents are lost when power is removed
- **RAM keeps data quickly accessible to the processor**
- **Every computing device has RAM: desktop, tablet, smartphone**

PART E

Processing Unit

The CPU, multicore chips and cache levels that do the computing.

Multi-core | Shared Bus to Memory | SIMD | Single Instruction Multiple Data

Computer Hardware

Hardware: What are they?



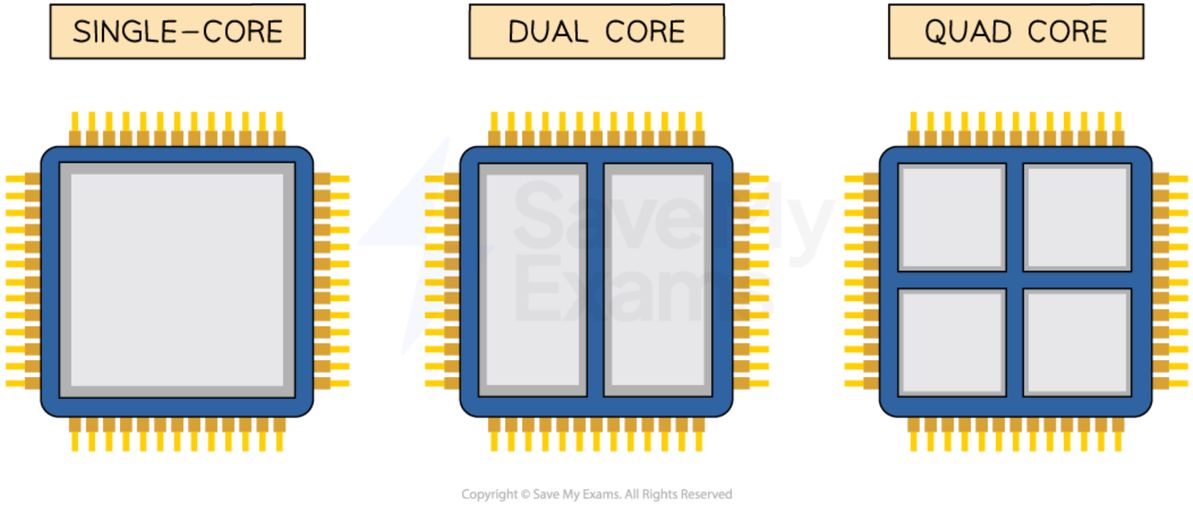




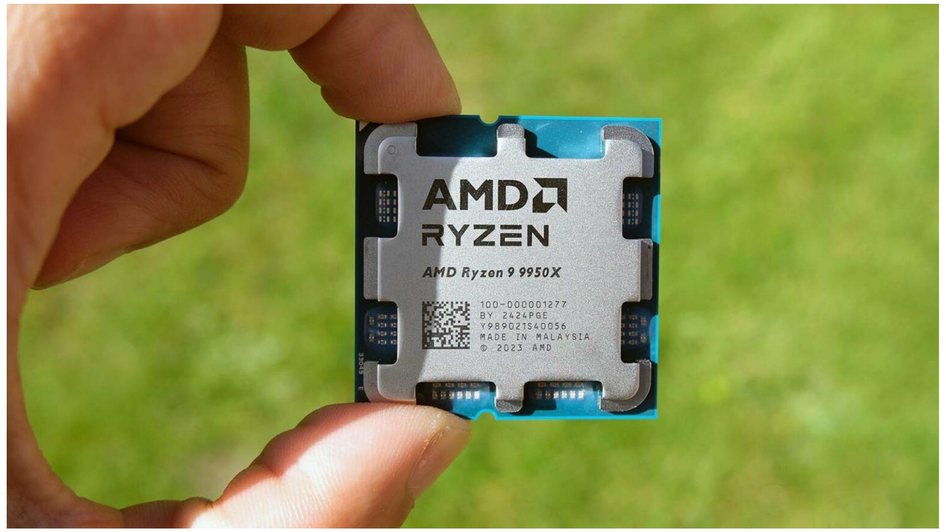
Computer Hardware

Hardware: Central Processing Unit

- **What are the famous manufacturers**?
- A chip/processor chip.
- The first single-core processor had one core only.
- Technological development has led to an increase in the number of cores.



Copyright © Save My Exams. All Rights Reserved



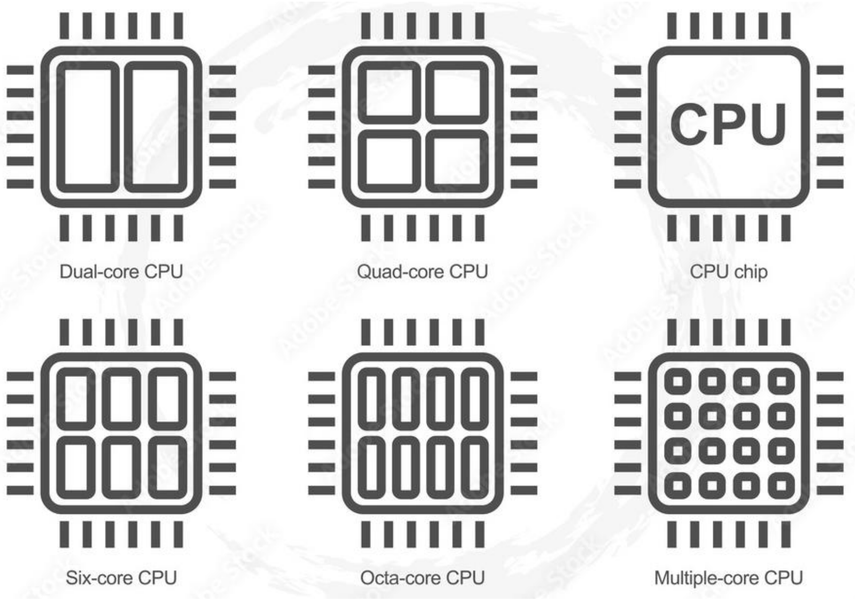
source: https://www.club386.com/heres-what-happens-when-you-run-an-amd-ryzen-9-9950x-at-65w/

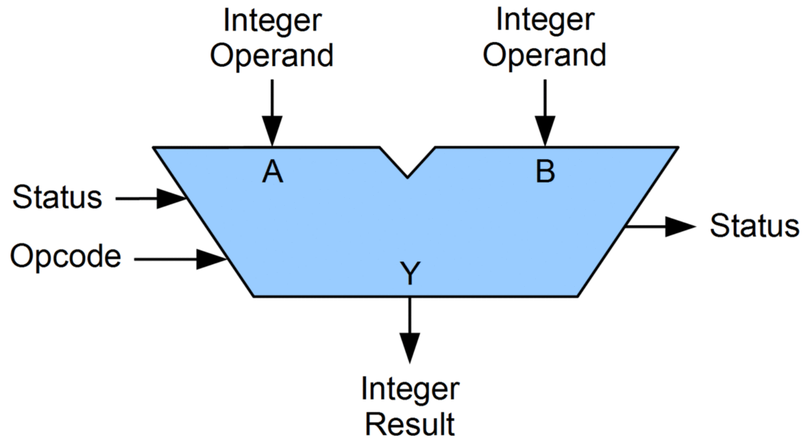
Computer Hardware

Hardware: Central Processing Unit

- A multicore computer, also known as a chip multiprocessor,
  - combines **two or more processors** (called cores) on a **single piece of** silicon (called a die).

- A processor, each of which contains its own
  - **control unit**,
  - arithmetic-logic unit (**ALU**), and
  - registers.

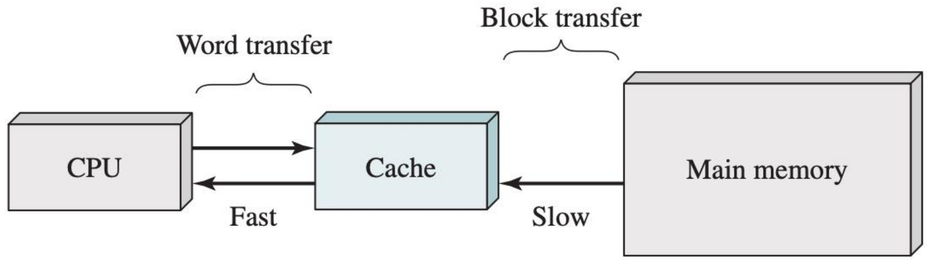




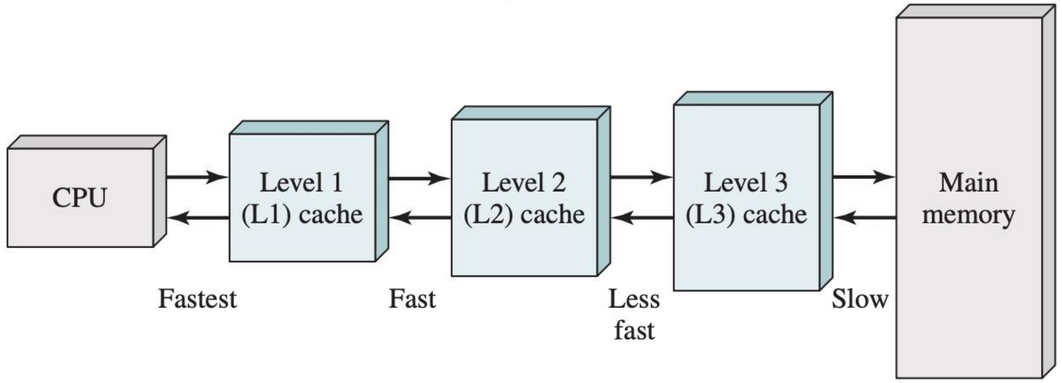
Computer Hardware

Hardware: Central Processing Unit

- Some processors typically have three dedicated **cache** levels: L1, L2, and L3.
- The figure indicates two different **cache**-level organizations between
  - one single **cache**, and
  - a three-level **cache**.



(a) Single cache



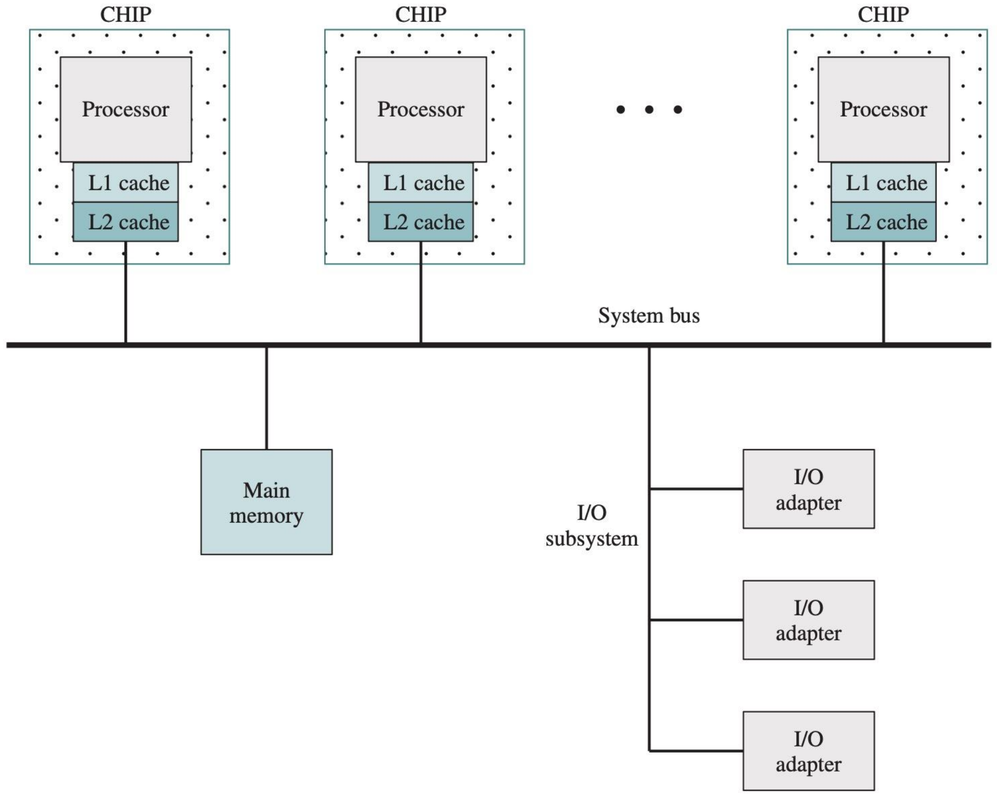
(b) Three-level cache organization

Computer Hardware

Hardware: Central Processing Unit

- Each processor typically has two dedicated levels of **cache**, designated L1 and L2.
- The figure indicates that **each processor and its dedicated caches** are housed on a separate chip.

- Each processor has access to a shared **main memory** and the I/O devices through some form of interconnection mechanism;
  - a **shared bus** is a **common** facility.



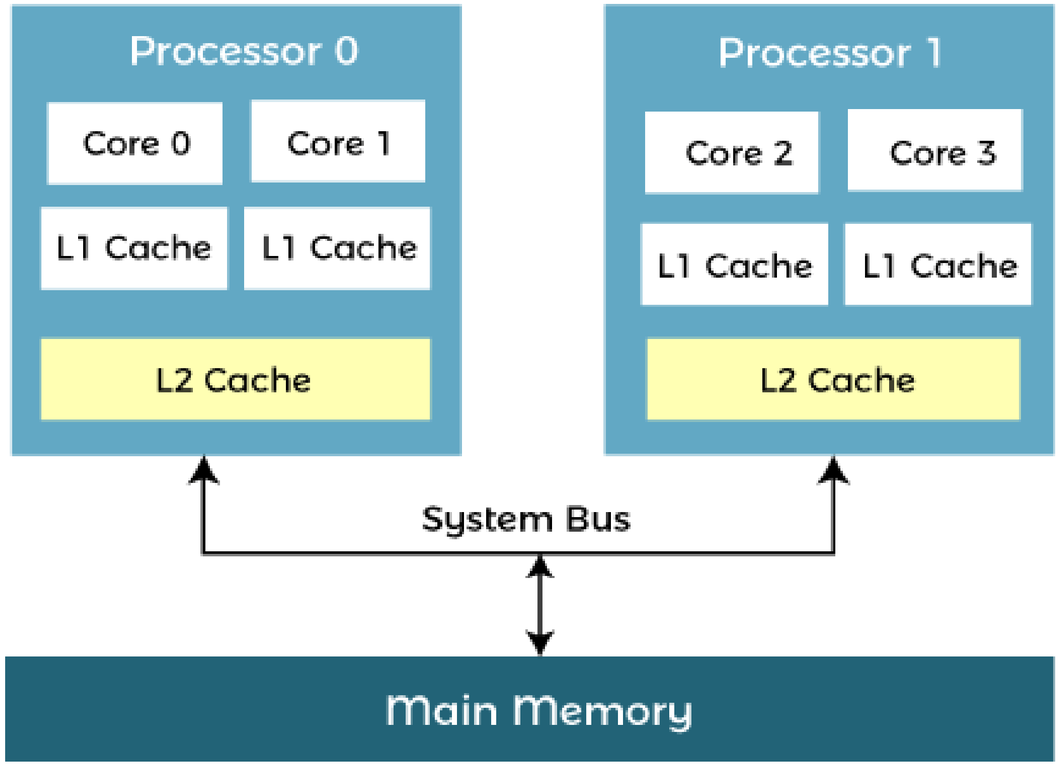
facility (n) - a feature of a product that allows you to do something.

Computer Hardware

Hardware: Central Processing Unit

- A quad-core processor has two dedicated levels of **cache**, designated L1 and L2 in each chip.
- The figure indicates that each processor and its dedicated caches are housed on a separate chip.

- Each processor has access to a shared **main memory** through some form of interconnection mechanism;
  - a **shared bus** called “System Bus” is a common **facility**.



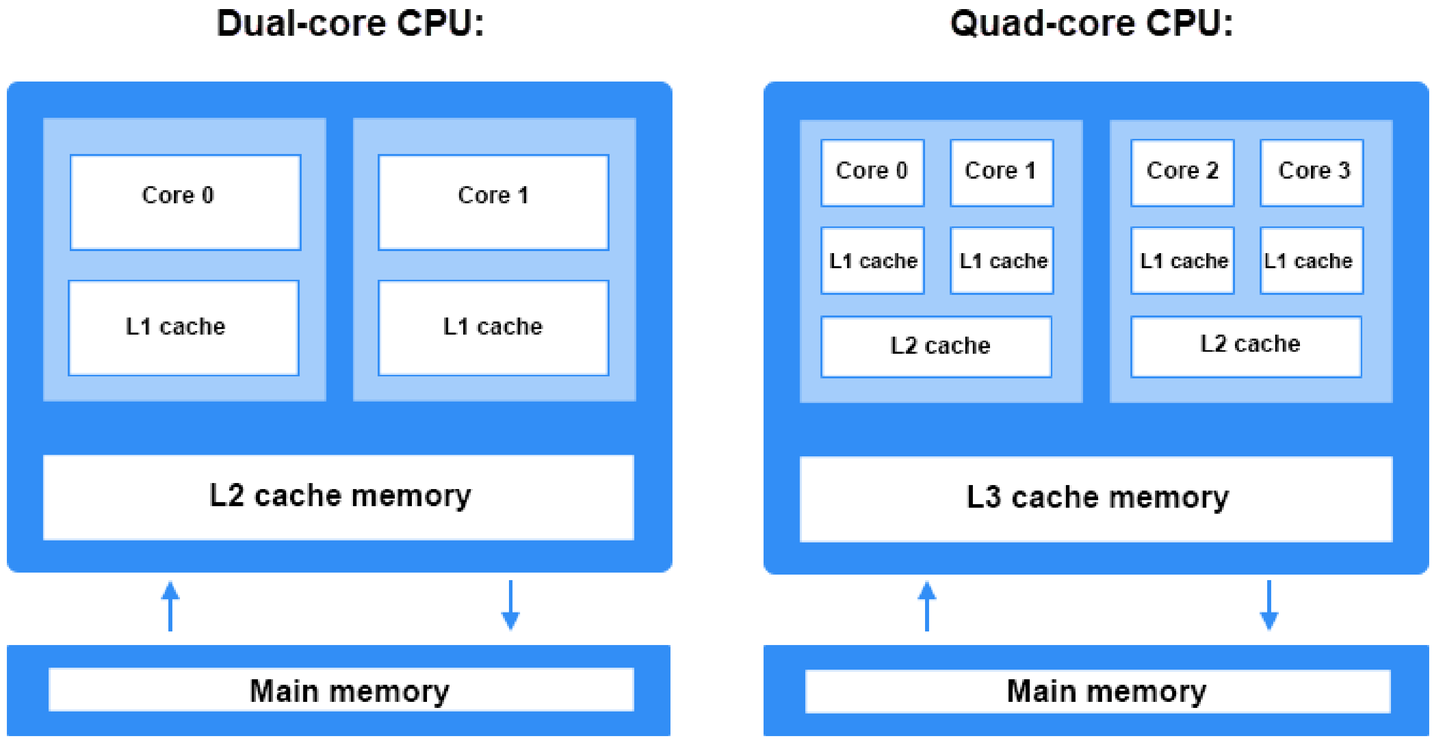
source: https://cpu392096892.wordpress.com/multi-core-cpus/

Computer Hardware

Hardware: Central Processing Unit

- A dual-core processor has two dedicated levels of **cache**, designated L1 and L2 in each chip.
- A quad-core processor has two dedicated levels of **cache**, designated L1 and L2 in each chip.

- Each processor has access to a shared **main memory** through some form of **interconnection mechanism** (**shared bus**).

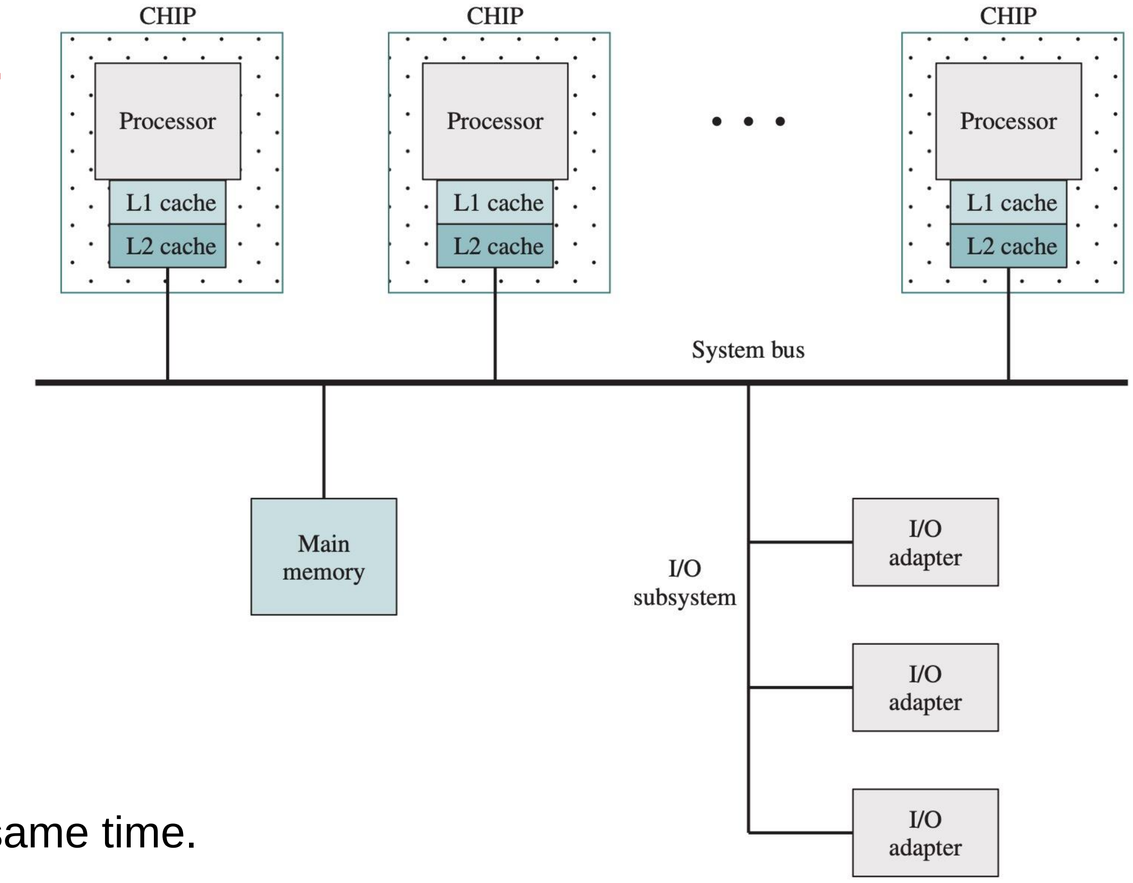


source: https://phoenixnap.com/kb/dual-core-vs-quad-core

Computer Hardware

Hardware: Central Processing Unit

- The **processors can communicate with each other through memory**
  - (messages and status information left in shared address spaces).
- It may also be possible for **processors to exchange signals** directly.
- The memory is often organized so that **multiple simultaneous accesses** to separate blocks of memory are possible.



simultaneous (adj) - happening or being done at exactly the same time.



Key Points: Processing Unit

- **The CPU controls the computer and performs data processing**
- **Multicore processors combine two or more cores on one chip**
- **Cache levels (L1, L2, L3) bridge the CPU-memory speed gap**
    - Cores communicate via shared memory or direct signal paths
- **The mechanism that let systems move from single-core to multi-core processors was the SIMD support.**
    - **Single Instruction Multiple Data. ➔ The shared bus.**

PART F

Computer Architectures Key Diagrams

Three essential diagrams: Von Neumann, the memory hierarchy and IEEE 754.

Computer Hardware

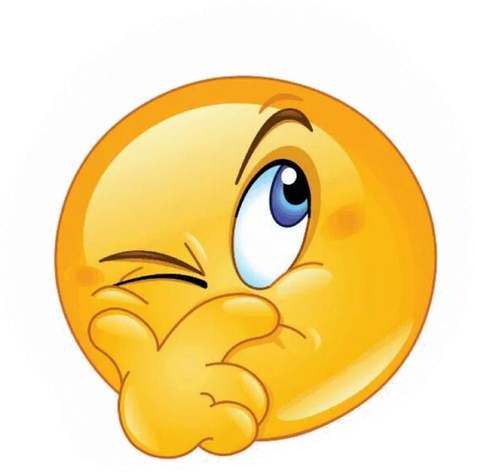
Hardware Designer

- thinks about
  - circuits,
  - components,
  - timing,
  - functionality,
  - ease of debugging

Computer Architecture

Architecture: What architecture will you design?

- . . .



Computer Architecture

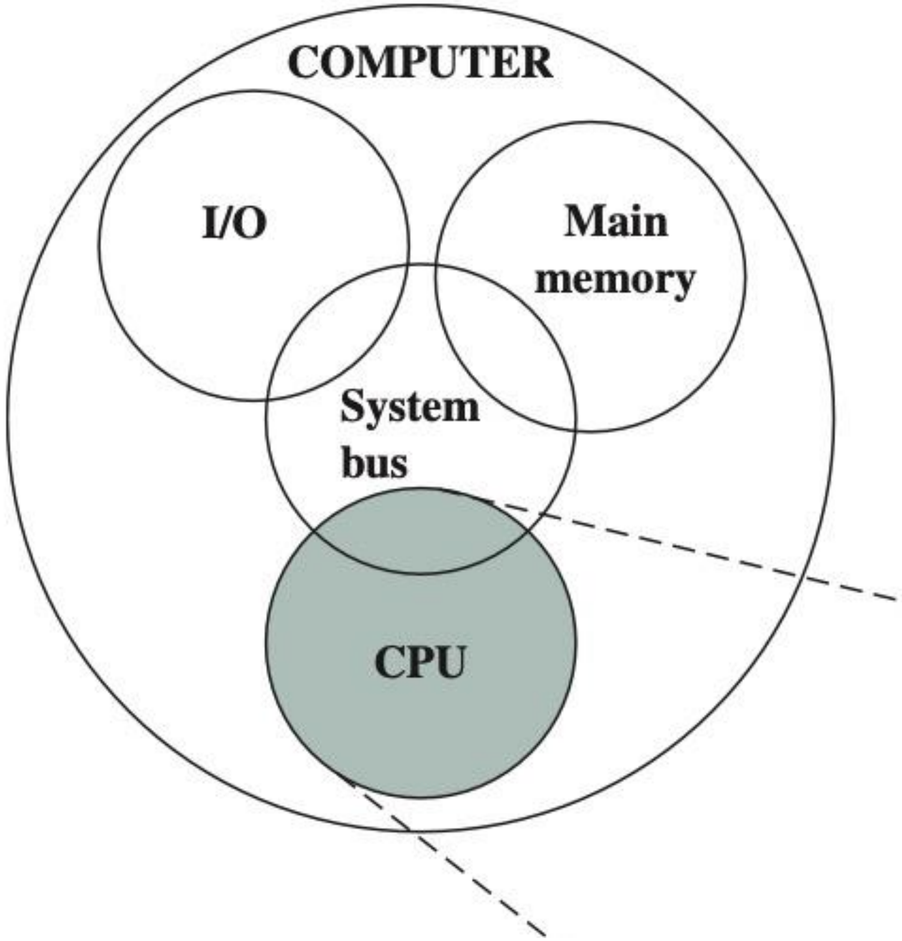
Computer Architect

- thinks about
  - high-level components,
  - how they fit together,
  - how they work together to deliver performance

Computer Top-Level Structure

Computer Organization and Architecture

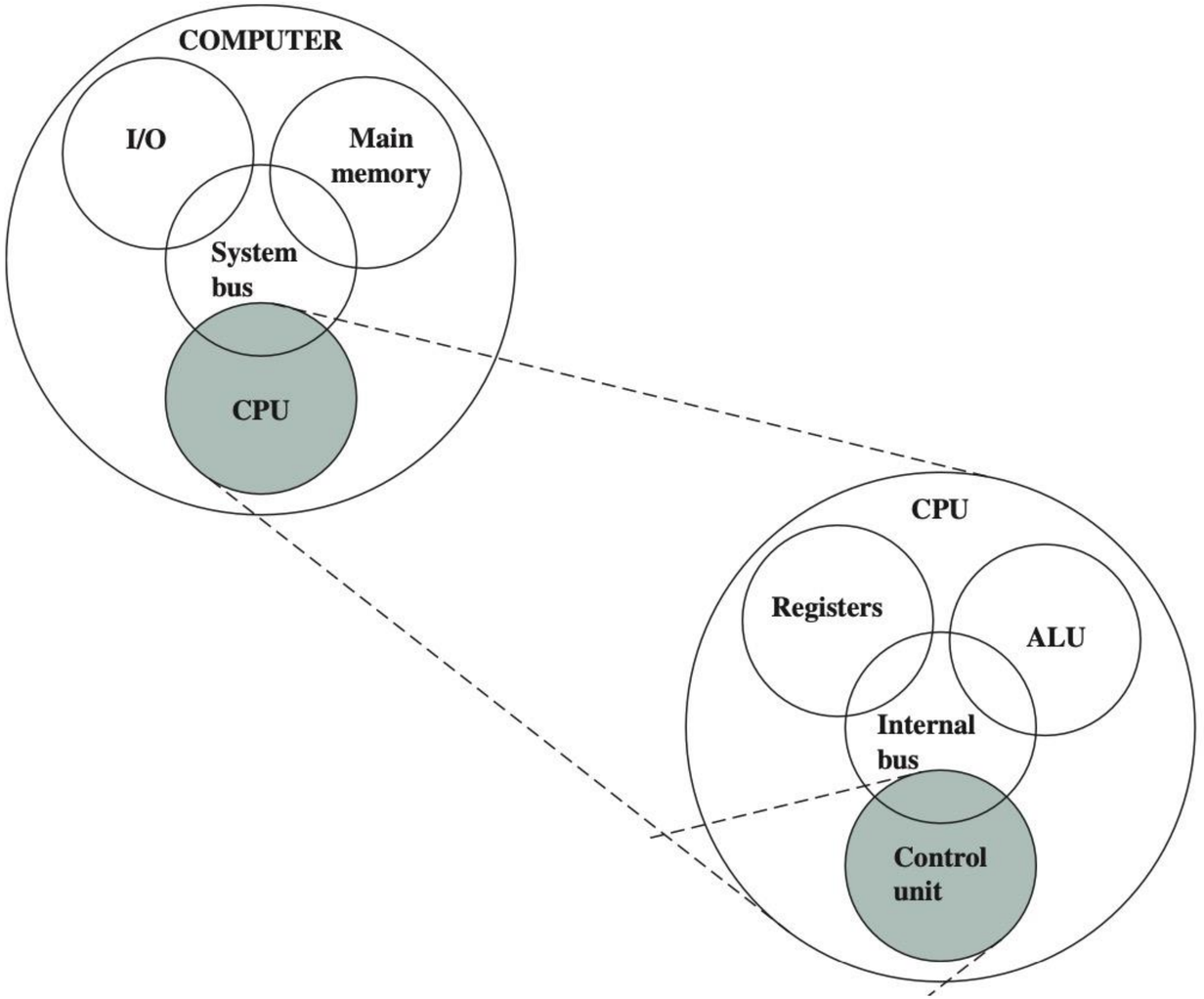
- thinks about
  - structure
  - function



Computer Top-Level Structure

Structure

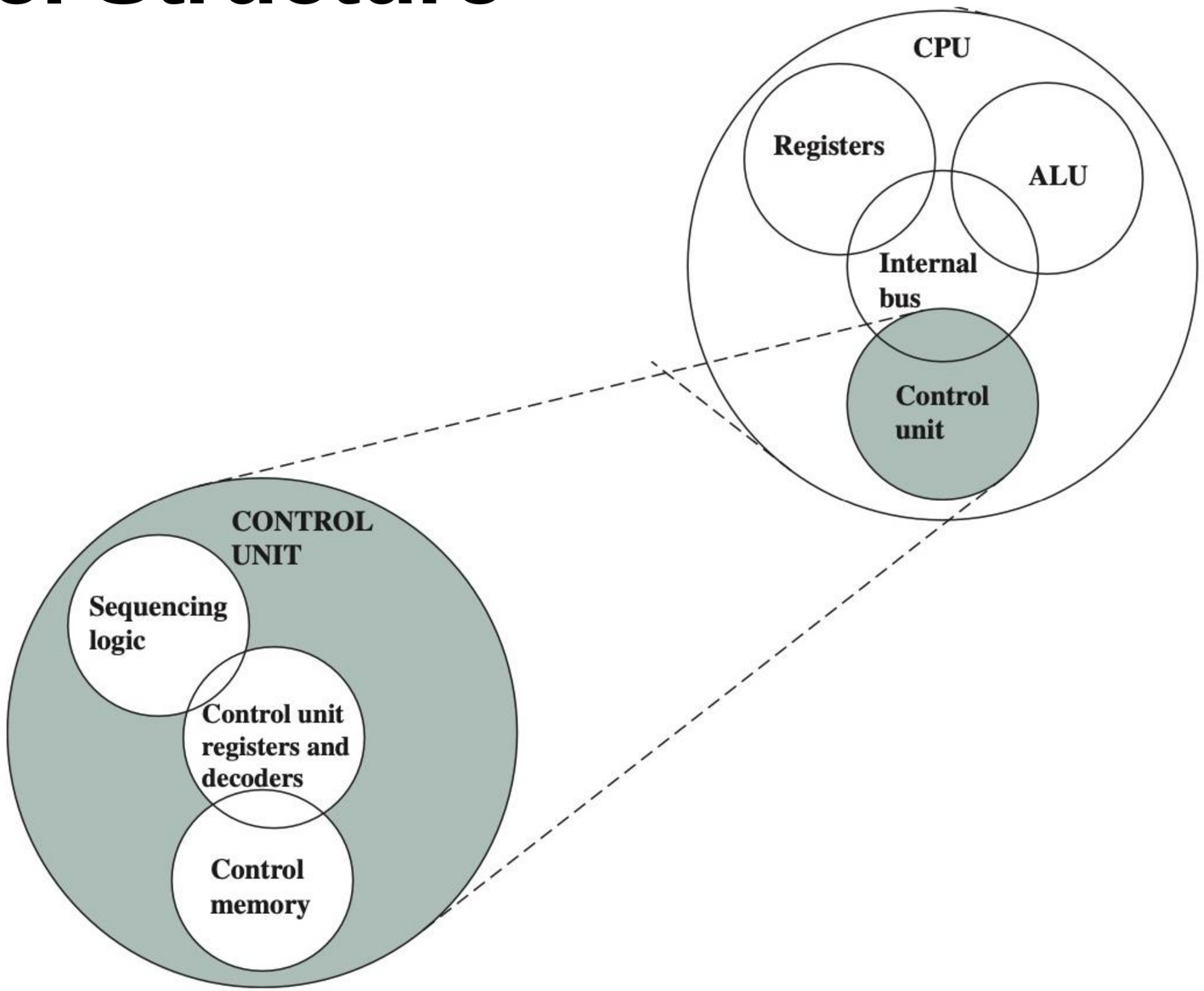
- Central processing unit (**CPU**):
  - Controls the operation of the computer,
  - Performs its data processing functions, and
  - Often referred to as the processor.
- **Main memory**:
  - Stores data.
- I/O:
  - Moves data between the computer and its external environment.



Computer Top-Level Structure

Structure

- System interconnection:
  - Some mechanism provides communication among the **CPU**, **main memory**, and I/O.
  - An example of system interconnection is a system bus, consisting of many conducting wires to which all the other components attach.



Computer Top-Level Structure

Three essential diagrams for your Computer Architecture

1. Von Neumann Architecture Diagram
2. **Memory Hierarchy** Pyramid
3. IEEE 754 Floating-Point Format (32-bit Single Precision)

Von Neumann Architecture Diagram

Von Neumann Architecture Diagram

- Structure:

```
[ Computer System ]
├── [ CPU ]
│    ├── [ ALU ]              ── Performs calculations
│    └── [ Control Unit ] ── Fetches & decodes instructions
├── [ Main Memory (RAM) ]
├── [ I/O Devices ]
└── [ Secondary Storage (HDD/SSD) ]
```

Von Neumann Architecture Diagram

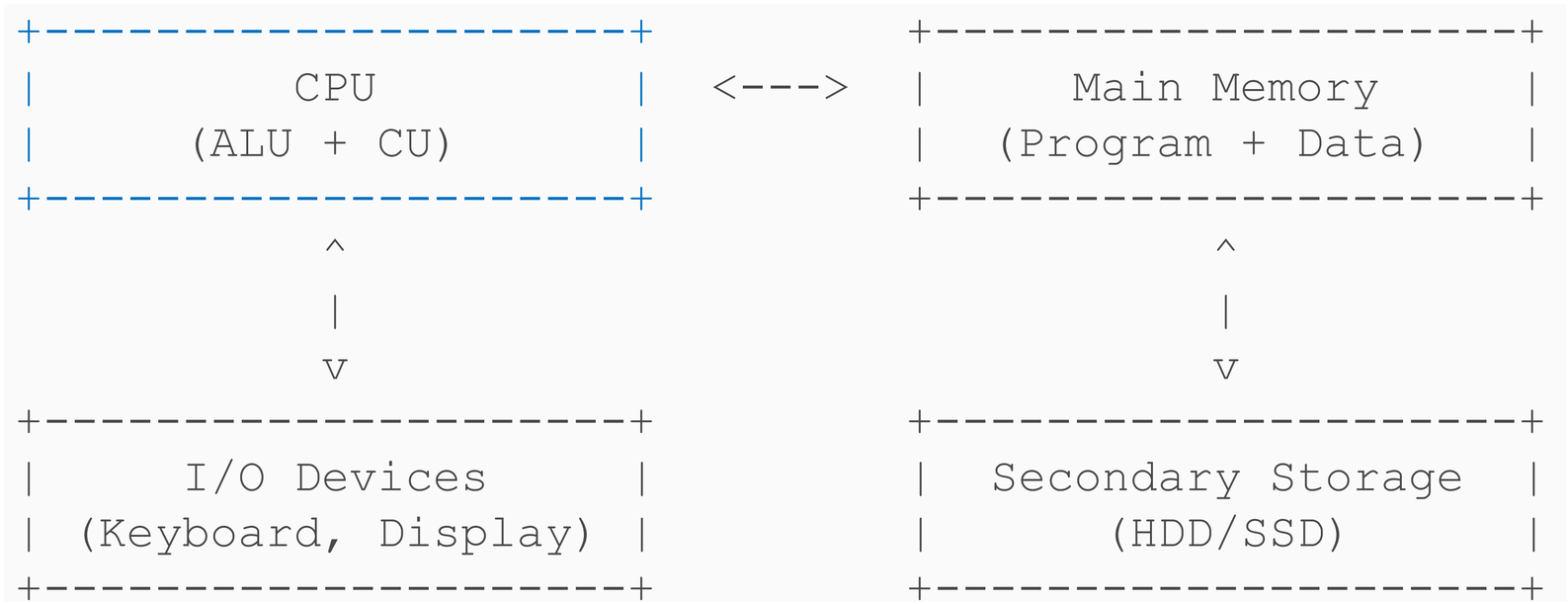
Von Neumann Architecture Diagram

- Key Features:
  - **Single shared bus** for data + instructions
  - Sequential execution (Fetch-Decode-Execute cycle)
  - Bottleneck: “Von Neumann bottleneck” due to its shared bus.

Von Neumann Architecture Diagram

Von Neumann Architecture Diagram

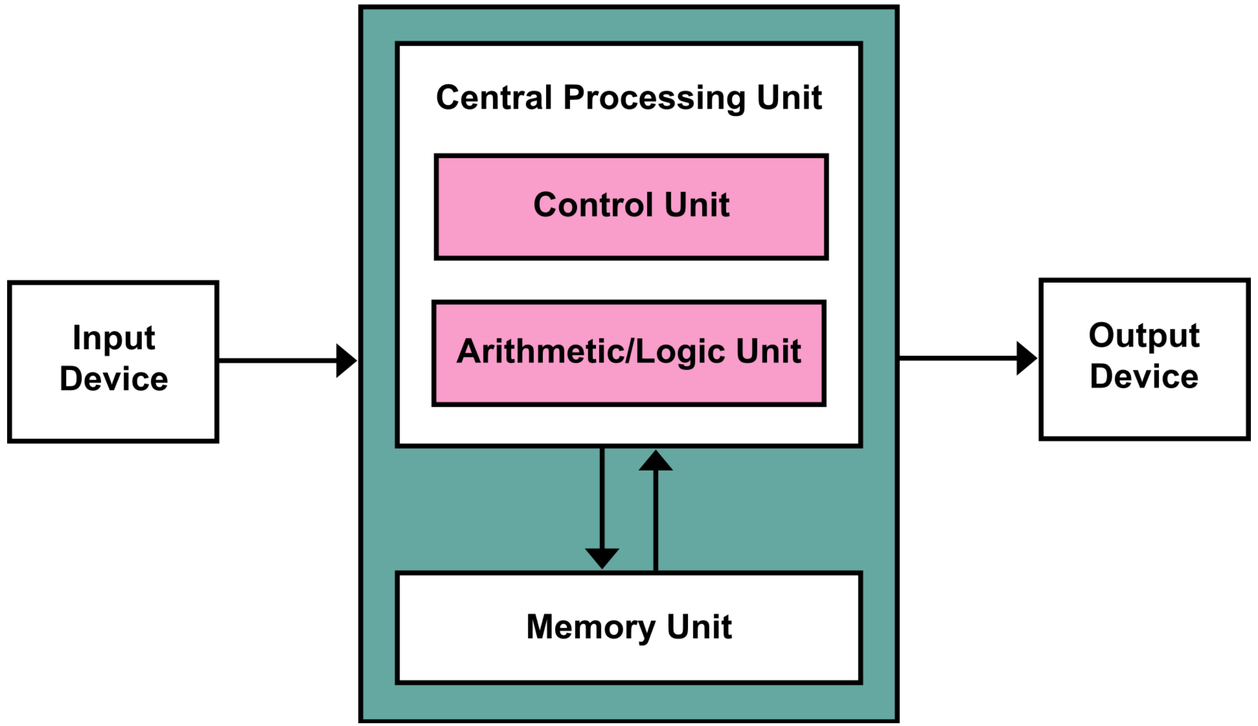
- Visual Representation:



Von Neumann Architecture Diagram

Von Neumann Architecture Diagram

- Visual Representation:

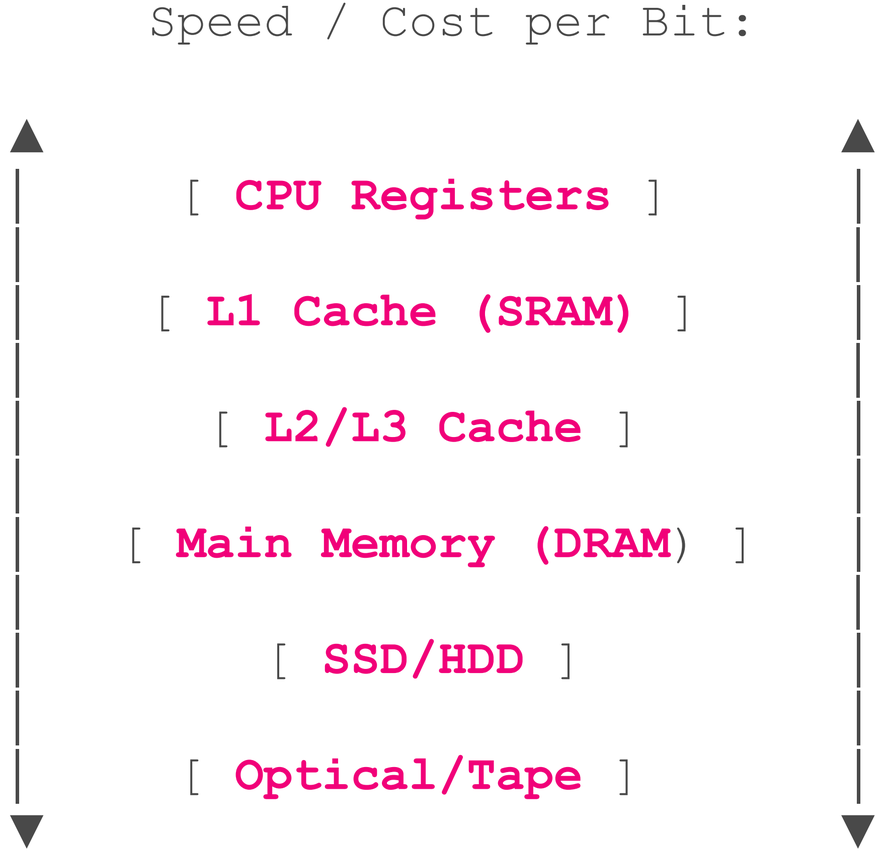


source: https://en.wikipedia.org/wiki/Von_Neumann_architecture#/media/File:Von_Neumann_Architecture.svg

Memory Hierarchy Pyramid

Memory Hierarchy Pyramid

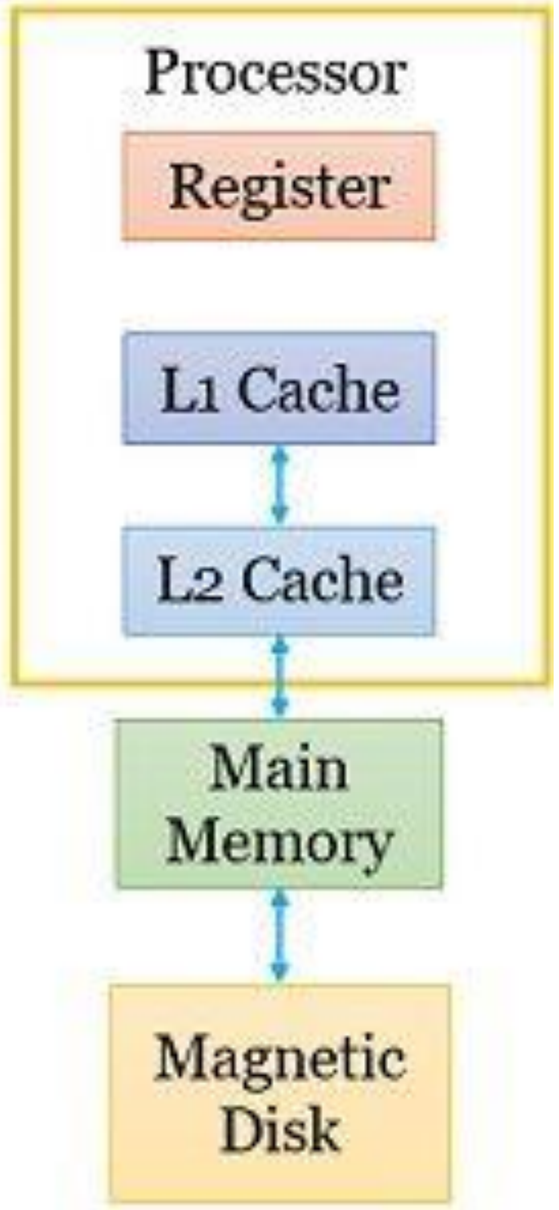
- Structure (Fastest → Slowest):



Memory Hierarchy Pyramid

Memory Hierarchy Pyramid

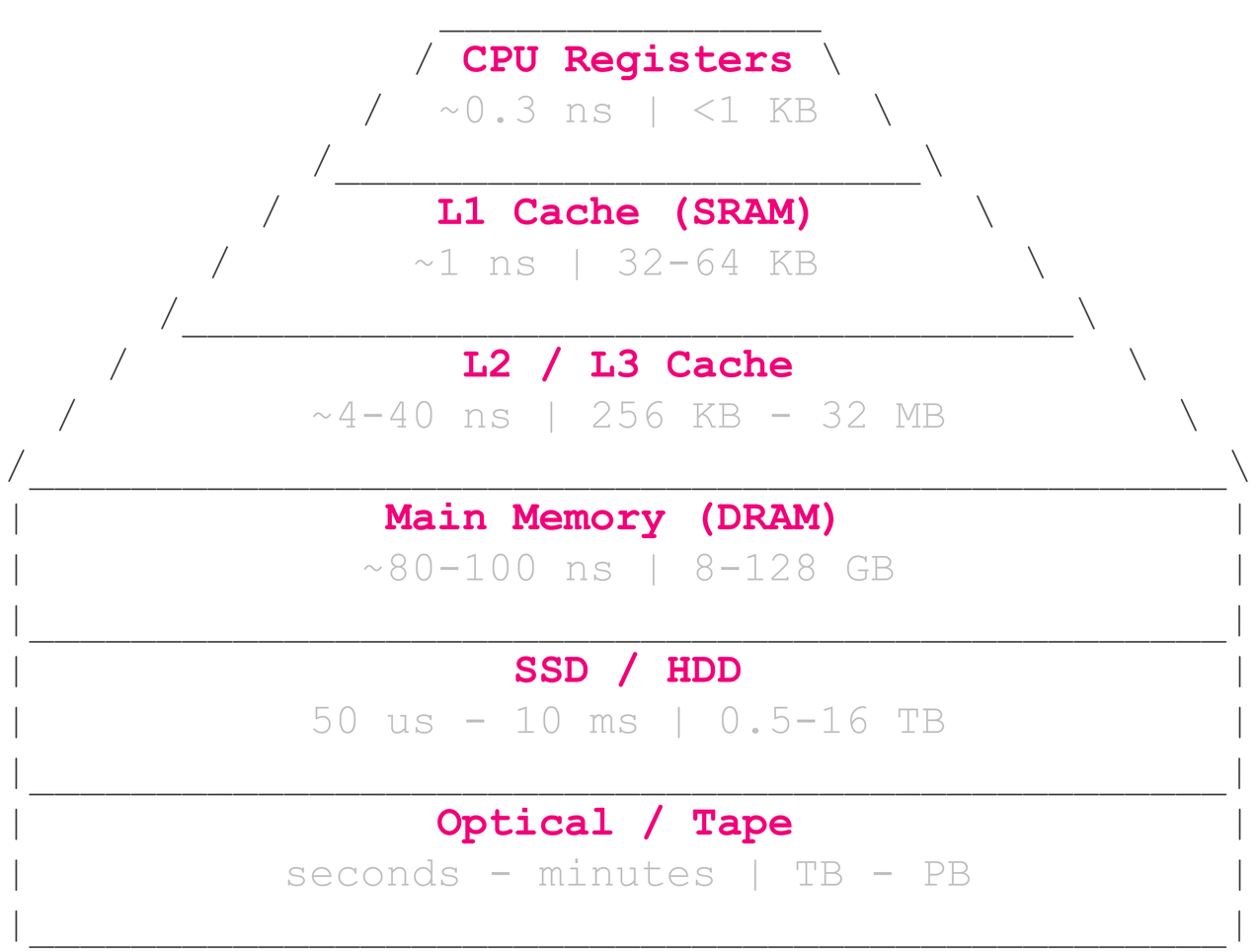
- Key Features:
  - **Registers**:
    - Fastest, most expensive (inside **CPU**)
  - **Cache** (L1/L2/L3):
    - SRAM, faster than DRAM
  - **Main Memory** (DRAM):
    - Volatile, slower than **cache**
  - Secondary Storage (**HDD** /**SSD**):
    - Non-volatile, slowest



Memory Hierarchy Pyramid

Memory Hierarchy Pyramid

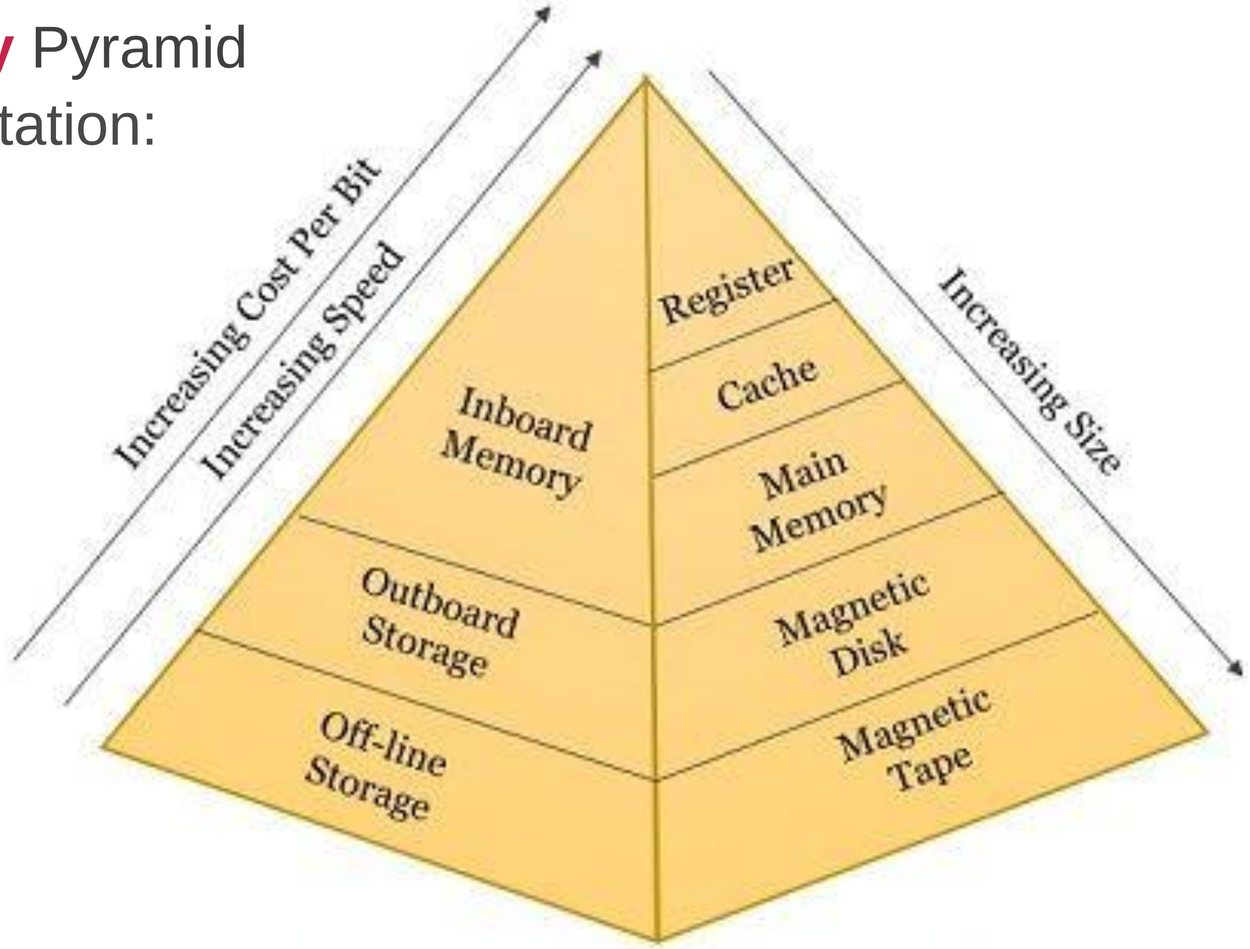
- Visual Representation:

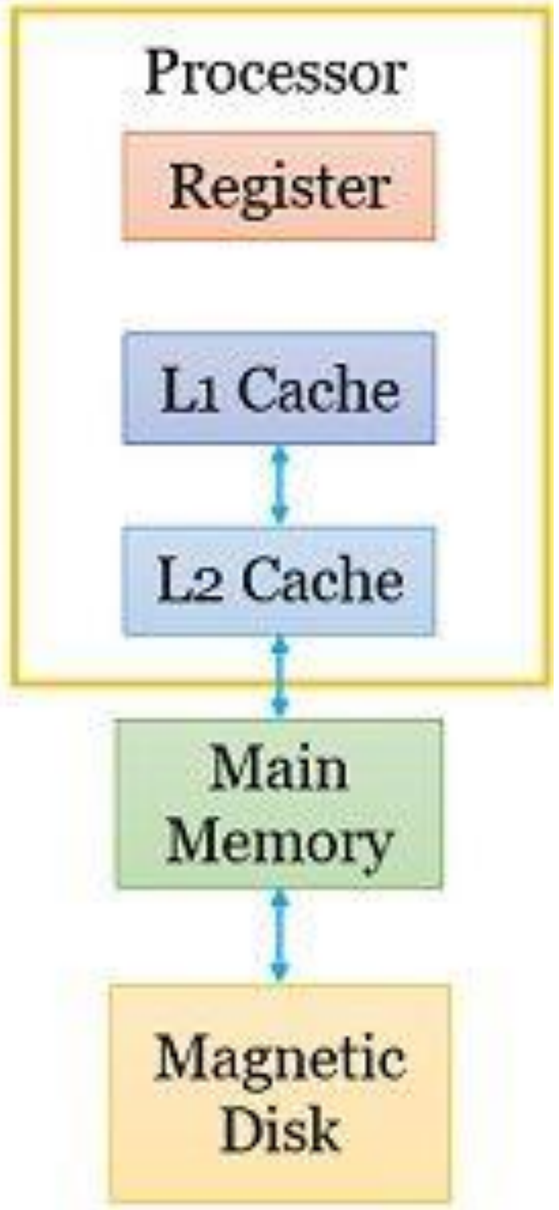


Memory Hierarchy Pyramid

Memory Hierarchy Pyramid

- Visual Representation:





source: https://binaryterms.com/memory-hierarchy-in-computer-architecture.html

PART F

Numbers into Computers

with the IEEE 754 Standard

IEEE 754 – Calculating and representing numbers in binary, a clever mechanism and interoperable standard for Computer Architectures.

The Problem with Floating Points:
Approximations of Decimals

Floating point numbers stored in binary are an **approximation** of decimal numbers.

The true value of 0.1:

```
>>> 0.1
0.1000000000000000055511151231257827021181583404541015625
```

```
from decimal import Decimal
>>> Decimal(0.1)
```

The displayed value by Python:

```
>>> 0.1
0.1
```

There's still a problem though, it's still an **approximation**:

```
>>> 0.1 + 0.2
0.30000000000000004
```

Open Python CLI:

0.3 – 0.2 – What is the output?

?

Binary Conversion (Base 2) Refresher

$5_{10} \rightarrow 101_2$

Binary Conversion

- Converting a decimal (whole) number to binary.

- Principle
  1. Take the decimal number as the numerator and divide it by 2.
  2. The remainder (0 or 1) is the least significant bit (LSB).
  3. Take the result from step 1 and divide it by 2.
  4. The remainder (0 or 1) is the next bit of the binary number.
  5. Repeat step 2 until you reach zero.
  6. The remainder is the most significant bit (MSB).

Binary Conversion (Base 2) Refresher

$5_{10} \rightarrow 101_2$

Binary Conversion

- Example
  - **Convert $(27)_{10}$ to binary.**

Least Significant Bit

2	27	remains	1 (LSB)
2	13	remains	1
2	6	remains	0
2	3	remains	1
2	1	remains	1
	0	remains	1 (MSB)

Most Significant Bit

  - The answer is $(11011)_2 = (27)_{10}$

Binary Conversion (Base 2) Refresher

$0.5_{10} \rightarrow 0.1_{2}$

Binary Conversion

- Converting a decimal number with a decimal point to binary.

- Principle
  1. Convert the integer in front of the decimal point using the previous method.
  2. Set the decimal point and multiply it by 2.
  3. If the product is less than 1, the binary value is 0.
  4. If the product is greater than 1 or equal to 1, the binary value is 1.
  5. Set the decimal point obtained from the multiplication in step 2, and multiply it by 2.
  6. Consider the result as in step 2.
  7. Continue this process until the product equals 1 or a sufficiently accurate value is obtained.

Binary Conversion (Base 2) Refresher

$0.5_{10} \rightarrow 0.1_{2}$

Binary Conversion

- Converting a decimal number to binary.
- Example
  - **Convert $(0.4375)_{10}$ to binary**.

Multiplication result	The result of integers
0.4375 x 2 = 0.875	0
0.8750 x 2 = 1.750	1
0.7500 x 2 = 1.500	1
0.5000 x 2 = 1.000	1

  - The answer is $(0.4375)_{10} = (0.0111)_{2}$.

Binary Conversion (Base 2) Refresher

$0.1_2 \rightarrow 0.5_{10}$

Binary Conversion

- Converting binary number to decimal numbers with decimal points
- Example
  - Convert $(0.0111)_2$ to a decimal number.

Base two	Base ten calculation
0.0111_2	$= (0x2^{-1}) + (1x2^{-2}) + (1x2^{-3}) + (1x2^{-4})$
	$= (0x0.5) + (1x0.25) + (1x0.125) + (1x0.0625)$
	$= (0) + (0.25) + (0.125) + (0.0625)$
	$= 0.4375$

  - The answer is $(0.0111)_2 = (0.4375)_{10}$

Binary Conversion (Base 2) Refresher

EXAM AND QUIZ PRACTICE

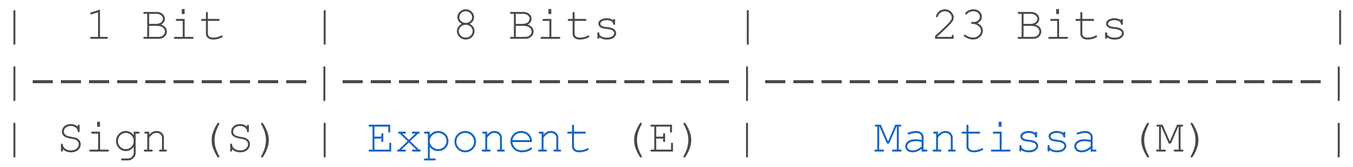
Binary Conversion

- Change $(12.35)_{10}$ to a binary number.
- Change $(103.92)_{10}$ to a binary number.
- Change $(101.1101)_{2}$ to a decimal number.
- Change $(1011.101)_{2}$ to a decimal number.

IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

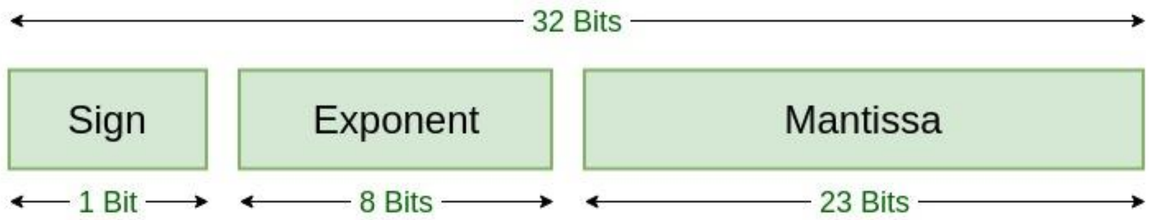
- Structure:



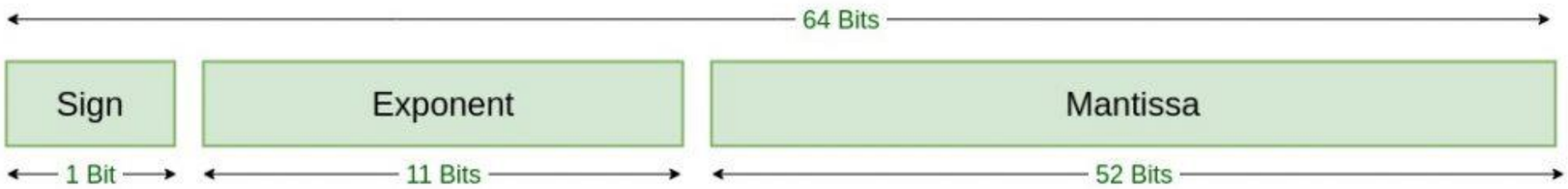
IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

- Visual Representation:



Single Precision
IEEE 754 Floating-Point Standard



Double Precision
IEEE 754 Floating-Point Standard

source: https://www.geeksforgeeks.org/computer-organization-architecture/ieee-standard-754-floating-point-numbers/

IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

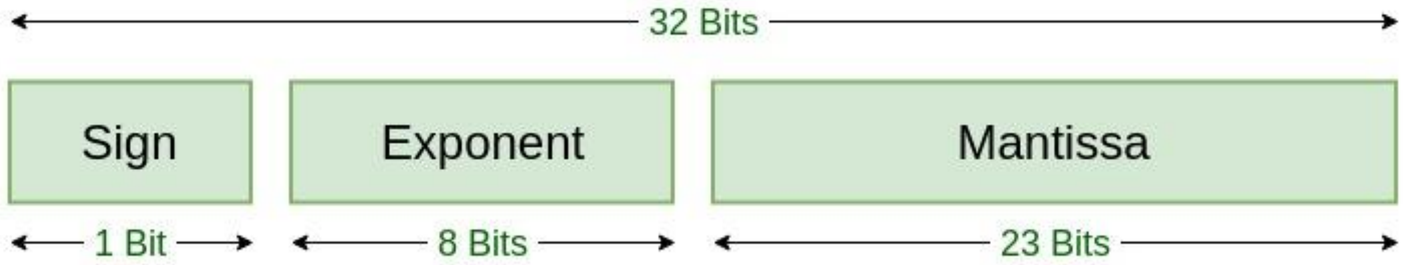
- It is a technical standard for floating-point computation that was established in 1985 by the Institute of Electrical and Electronics Engineers (IEEE).
- The standard addressed many problems found in the diverse floating point implementations that made them difficult to use reliably and reduced their portability.
- IEEE Standard 754 floating point is the most common representation today for real numbers on computers, including
  - Intel-based PC's,
  - Macs, and
  - most Unix platforms.

IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

- There are several ways to represent floating-point numbers, but IEEE 754 is the most common and is efficient in most cases.

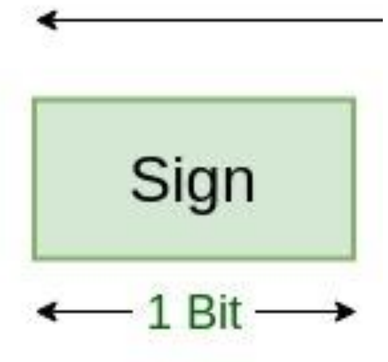
- IEEE 754 has 3 basic components:
  - The Sign of Mantissa
  - The **Biased** Exponent
  - The **Normalized** Mantissa



IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

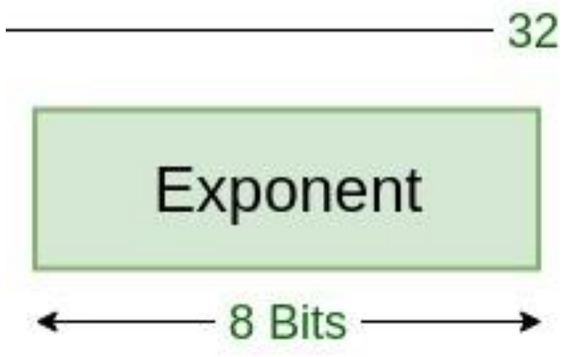
- The Sign of Mantissa
  - This is as simple as the name.
  - 0 represents a positive number, while 1 represents a negative number.



IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

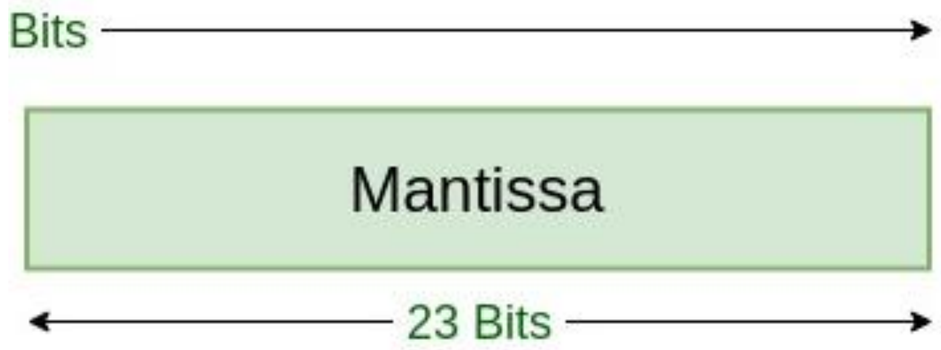
- The Biased exponent
  - The exponent field needs to represent both positive and negative exponents.
  - A bias is added to the actual exponent in order to get the stored exponent.



IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

- The **Normalised** Mantissa
  - The mantissa is part of a number in scientific notation or a floating-point number, consisting of its significant digits.
  - Here we have only 2 digits, i.e., 0 and 1.
  - So a **normalised** mantissa is one with **only one 1 to the left** of the decimal.

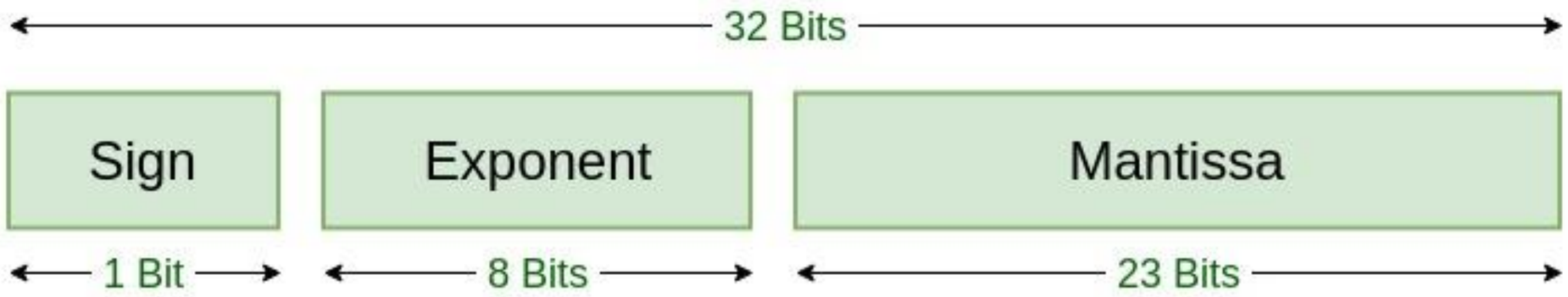


IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

- Visual Representation:

```
+-------+--------------+--------------------------+
|   1   |   10000010   | 10110100000000000000000  |
+-------+--------------+--------------------------+
|  Sign |   Exponent   | Mantissa                 |
+-------+--------------+--------------------------+
```



IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

- Key Features:
  - Sign (S): 0 = Positive, 1 = Negative
  - Exponent (E): Bias = 127 (Stored as E + 127)
  - Mantissa (M): *Implicit leading 1 (Normalized)*

What is the binary exponent, if the IEEE 754 the exponent is 125 ?
127 ?
126 ?

Why?: Designed for numbers with negative exponents to be **compared** with subtraction (fast) instead of more complicated circuitry.

IEEE 754 Floating-Point Format

WORKED EXAMPLE

IEEE 754 Floating-Point Format (32-bit Single Precision)

- Example: 85.125

Integer part (85):

85 = 1010101

Fractional part :

0.125 = 001

Combined binary (85.125):

85.125 = 1010101.001

IEEE 754 Floating-Point Format

WORKED EXAMPLE

IEEE 754 Floating-Point Format (32-bit Single Precision)

- Example: 85.125

Normalize the Binary Number

85.125 = 1010101.001

= 1.010101001 x 2^6 ← normalization to "1.xxxxxxx"

Sign bit: 0 (positive number).

Exponent: 6 (since we shifted the binary point 6 places left).

Mantissa: 010101001 (the fractional part after normalization). *<Implicit **1.** is assumed>*

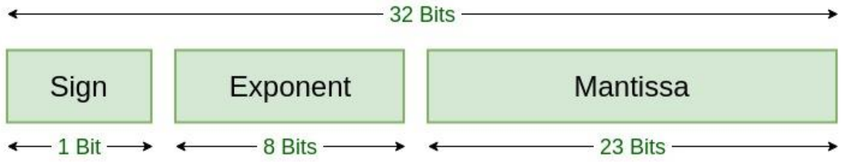
IEEE 754 Floating-Point Format

WORKED EXAMPLE

IEEE 754 Floating-Point Format (32-bit Single Precision)

- Example: 85.125 on IEEE 754 Single-Precision (32-bit)

Field	Size (bits)	Calculation	Binary Value
Sign (S)	1	0 (positive)	0
Exponent (E)	8	127 (bias) + 6 = 133	10000101 (133 in binary)
Mantissa (M)	23	010101001 (padded with zeros)	01010100100000000000000



Single Precision
IEEE 754 Floating-Point Standard

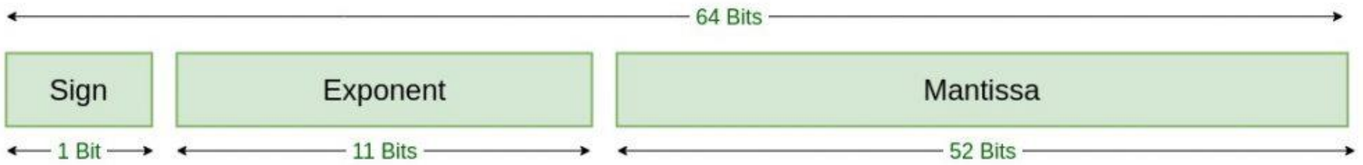
IEEE 754 Floating-Point Format

WORKED EXAMPLE

IEEE 754 Floating-Point Format (64-bit Single Precision)

- Example: 85.125 on IEEE 754 Single-Precision (64-bit)

Field	Size (bits)	Calculation	Binary Value
Sign (S)	1	0 (positive)	0
Exponent (E)	11	1023 (bias) + 6 = 1029	10000000101 (1029 in binary)
Mantissa (M)	52	010101001 (padded with zeros)	01010100100...00(52 bits)



Double Precision
IEEE 754 Floating-Point Standard



Key Points: Computer Architectures (Key Diagrams)

- **Structure describes how components relate; function is what they do**
- **Von Neumann: single shared bus for data and instructions**
  - Fetch-Decode-Execute cycle; shared bus causes the bottleneck
- **Memory hierarchy trades speed vs. cost per bit at every level**
- **IEEE 754 floats: sign + biased exponent + normalized mantissa**
  - Single precision uses bias 127: value = 1.xxxx × 2^(E-127)

IEEE 754 Floating-Point Format - Key Takeaways

EXAM AND QUIZ PRACTICE

IEEE 754 Floating-Point Format — Key Takeaways:

- **Normalization:** The binary number is expressed as $1.xxxx \times 2^n$
- Exponent Bias:
  - **Single**-precision: Add **127** to the exponent.
  - **Double**-precision: Add **1023** to the exponent.
- Mantissa Padding:
  - The mantissa **is padded with trailing zeros** to fill the required bits (23 for single, 52 for double).
- This ensures the number is stored efficiently in IEEE 754 format.

Uses of These Diagrams - Key Takeaways

EXAM AND QUIZ PRACTICE

Uses of These Diagrams:

- **Von Neumann**: Explain the bottleneck and unified memory.
- **Memory Hierarchy**: Discuss trade-offs (speed vs. cost).
- **IEEE 754**: Walk through the example of -13.625.

EXAM AND QUIZ PRACTICE

Uses of These Diagrams

Quiz on MyCourses

- **IEEE 754**: Walk through the floating point value of … .



References

References

Reference Materials

- William Stallings, “Computer Organization and Architecture Designing for Performance,” Tenth Edition, *Pearson*, 2016
- React, “Kids react to old computers,” https://www.youtube.com/watch?v=PF7EpEnglgk
- Maja Olejniczak, “English for Information Technology 1,” Series editor David Bonamy, *Pearson Longman*, UK, 2011
- http://ecusbinchen.blogspot.com/2010/11/system-component.html
- https://blog.kiatoo.com/conseils-achat/hdd-vs-sshd-vs-ssd-differences-2984

Mahidol University

IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

- To Represent Signed Exponents
  - The exponent can be positive or negative
    - e.g., 2^{6} or 2^{-3}
  - Instead of using two's complement (which complicates hardware design), IEEE 754 uses a biased exponent (always stored as a positive number).

Fixed Exponent Bias Values

Precision	Exponent Bits	Bias Value	Formula for Bias
Single (32-bit)	8 bits	**127**	$2^{(8-1)} - 1 = 127$
Double (64-bit)	11 bits	**1023**	$2^{(11-1)} - 1 = 1023$

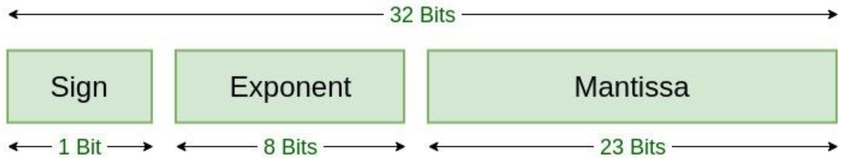
IEEE 754 Floating-Point Format

IEEE 754 Floating-Point Format (32-bit Single Precision)

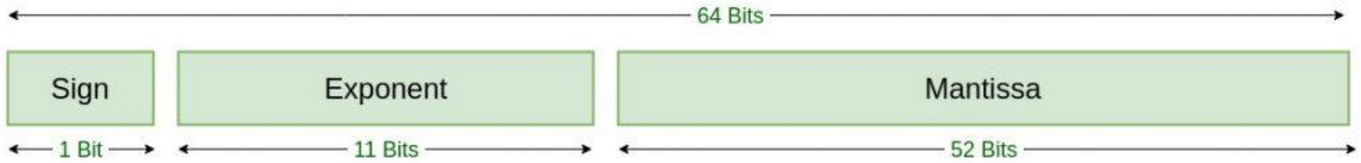
- Fixed Exponent Bias Values

Fixed Exponent Bias Values

Precision	Exponent Bits	Bias Value	Formula for Bias
Single (32-bit)	8 bits	**127**	$2^{(8-1)} - 1 = 127$
Double (64-bit)	11 bits	**1023**	$2^{(11-1)} - 1 = 1023$



Single Precision
IEEE 754 Floating-Point Standard



Double Precision
IEEE 754 Floating-Point Standard